

What is it?

Arts participation is defined as involvement in artistic and creative activities, such as dance, drama, music, painting, or sculpture. It can occur either as part of the curriculum or as extra-curricular activity. Arts-based approaches may be used in other areas of the curriculum, such as the use of drama to develop engagement and oral language before a writing task.

Participation may be organised as regular weekly or monthly activities, or more intensive programmes such as summer schools or residential courses. Whilst these activities, of course, have important educational value in themselves, this Toolkit entry focuses on the benefits of Arts participation for core academic attainment in other areas of the curriculum particularly literacy and mathematics.

Key findings

1. Arts participation approaches can have a positive impact on academic outcomes in other areas of the curriculum.
2. The research here summarises the impact of arts participation on academic outcomes. It is important to remember that arts engagement is valuable in and of itself and that the value of arts participation should be considered beyond maths or English outcomes.
3. If the aim of the arts approach is to improve academic attainment it is important to identify the link between your chosen arts intervention and the outcomes you want to improve.
4. Arts-based approaches may offer a route to re-engage older pupils in learning, though this does not always translate into better attainment. It is important to consider how you will use increased engagement to improve teaching and learning for these pupils.

How effective is the approach?

Overall, the average impact of arts participation on other areas of academic learning appears to be positive but moderate, about an additional three months progress.

Improved outcomes have been identified in English, mathematics and science. Benefits have been found in both primary and secondary schools.

Some arts activities have been linked with improvements in specific outcomes. For example, there is some evidence of the impact of drama on writing and potential link between music and spatial awareness.

Wider benefits such as more positive attitudes to learning and increased well-being have also consistently been reported.

Behind the average

Closing the disadvantage gap

There is intrinsic value in teaching pupils creative and performance skills and ensuring disadvantaged pupils access a rich and stimulating arts education. Arts participation may be delivered within the core curriculum, or through extra-curricular or cultural trips which can be subject to financial barriers for pupils from deprived backgrounds.

There is some evidence to suggest a causal link between arts education and the use of arts-based approaches with overall educational attainment. Where the arts are being taught as a means to boost academic achievement for those eligible for the pupil premium, schools should carefully monitor whether this aim is being achieved.

How could you implement in your setting?

Arts participation relates to a broad range of subjects including traditional fine arts, theatre, dance, poetry, and creative writing. It also includes teaching strategies which explicitly include arts elements, such as drama-based pedagogy.

Some components of arts education approaches might include:

Arts education may be organised as regular lessons or monthly activities, after school clubs, small group or one-on-one tuition or

whole school approaches. Activities can also be delivered through more intensive programmes such as summer schools or residential courses.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School's Guide to Implementation](#).

What does it cost

The average cost of arts education is expected to be very low, with costs ranging from very low to high depending on the type of provision. Costs to schools are largely based on teacher professional development and resources. Costs are greater where activities fall outside of the school day or involve small group or 1:1 tuition from specialist teachers.

Implementing arts education will require a small amount of additional staff time compared with other approaches as it is part of the core curriculum. Arts activities may also involve professional artists, certified drama or music teachers.

Alongside time and cost, school leaders should consider how to maximise professional development needs of staff to effectively integrate arts activities (such as drama, visual arts or music) in the classroom and evaluate their impact on pupil outcomes.

How secure is the evidence?

The security of the evidence around Arts Participation is rated as moderate. 80 studies were identified. The topic lost a padlock because a large percentage of the studies were not independently evaluated. Evaluations conducted by organisations connected with the approach – for example, commercial providers, typically have larger impacts, which may influence the overall impact.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

Evidence strength

Number of studies

What is it?

By aspirations we mean the things children and young people hope to achieve for themselves in the future. To meet their aspirations about careers, university, and further education, pupils often require good educational outcomes.

Raising aspirations is therefore often believed to incentivise improved attainment.

Aspiration interventions tend to fall into three broad categories:

1. interventions that focus on parents and families;
2. interventions that focus on teaching practice; and
3. out-of-school interventions or extra-curricular activities, sometimes involving peers or mentors.

The approaches used in these interventions are diverse. Some aim to change aspirations directly by exposing children to new opportunities and others aim to raise aspirations by developing general self-esteem, motivation, or self-efficacy. For interventions that focus on self-efficacy and motivation specifically in a learning context (for example, growth mindsets interventions) please see [Metacognition and self-regulation](#).

Key findings

1. The current evidence base on aspiration interventions is extremely weak. The lack of studies identified means that an impact in months progress is not communicated. Schools should carefully monitor the impact on attainment of any interventions or approaches.

2. The existing wider evidence suggests that the relationship between aspirations and attainment is not straightforward. In general, approaches to raising aspirations have not translated into increased learning. Approaches linked to gains in attainment almost always have a significant academic component,

suggesting that raising aspirations in isolation will not be effective

3. Most young people have high aspirations for themselves.

Ensuring that students have the knowledge and skills to progress towards their aspirations is likely to be more effective than intervening to change the aspirations themselves.

4. The attitudes, beliefs, and behaviours that surround aspirations in disadvantaged communities are diverse, so avoid generalisations.

How effective is the approach?

The lack of studies identified that tested aspiration interventions mean that there is not enough security to communicate a month's progress figure.

It is important to acknowledge that wider evidence indicates that the relationship between aspirations and attainment is complex, and there are many reasons why aspiration interventions may or may not impact upon attainment.

Some studies have shown that most young people already have high aspirations, suggesting that much underachievement results not from low aspiration but from a gap between aspirations and the knowledge, skills, and characteristics required to achieve them.

Where pupils do have lower aspirations, it is not clear whether targeted interventions have consistently succeeded in raising their aspirations. Also, where aspirations begin low and are successfully raised by an intervention, it is not clear that an improvement in learning necessarily follows.

Closing the disadvantage gap

Whilst pupils eligible for the pupil premium are likely to have lower academic attainment compared to their more advantage peers, the assumption that poorer pupils have lower aspirations for their education and adult life is less clear.

Studies in England suggest that different socioeconomic groups have similar levels of aspiration for their future outcomes and that differences in participation rates in higher education are largely driven by academic attainment. Given the broad range of attitudes, behaviours and beliefs surrounding aspirations in communities with higher rates of poverty, schools should avoid generalisations.

Aspiration interventions without an academic component are unlikely to narrow the disadvantaged attainment gap. Teacher expectations play a role in shaping pupil outcomes and teachers should aim to communicate a belief in the academic potential of all pupils.

How could you implement in your setting?

Aspiration approaches are diverse and may focus on parents and families, teaching practice or out-of-school interventions or extra-curricular activities involving peers or mentors. When implementing aspiration interventions, schools might consider including:

Given the limited evidence base, it is particularly important to monitor the impacts where aspiration approaches are used as a method of improving attainment.

Aspiration interventions range in duration and may include within class approaches delivered by teachers, after school clubs, out-of-school programmes, or mentoring led by paid staff or volunteers. Mentoring and parental interventions are typically delivered over an extended period of time (often at least the length of an academic year) in order to build effective relationships.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School’s Guide to Implementation](#).

What does it cost

Costs vary widely and are hard to estimate precisely, but overall, they are estimated to range between very low and moderate depending on the approach. After school clubs typically costs about £5 to £10 per session, so a weekly programme lasting 20 weeks might cost up to £200 per pupil. The median costs of implementing parental engagement programmes are estimated as very low to moderate, with greater costs where schools cover additional staffing costs.

The median cost of a mentoring intervention is estimated as moderate. The costs to schools are largely based on mentor training, salary costs (for non-volunteer mentors) and resources. Some programmes also include continuous training and support for mentors which may increase costs.

Alongside time and costs, school leaders should consider how to maximise the effectiveness of approaches by including a significant academic component and avoid approaches that aim to raise aspirations in isolation which may not be effective.

How secure is the evidence?

The security of the evidence around aspiration interventions is rated as extremely low. For topics with extremely low evidence, a month’s progress figure is not displayed. Only 3 studies were identified that met the pre-specified inclusion criteria.

What is it?

Behaviour interventions seek to improve attainment by reducing challenging behaviour in school. This entry covers interventions aimed at reducing a variety of behaviours, from low-level disruption to aggression, violence, bullying, substance abuse and general anti-social activities. The interventions themselves can be split into three broad categories:

1. Approaches to developing a positive school ethos or improving discipline across the whole school which also aim to support greater engagement in learning;
2. Universal programmes which seek to improve behaviour and generally take place in the classroom; and
3. More specialised programmes which are targeted at students with specific behavioural issues.

Other approaches, such as Parental engagement and Social and emotional learning programmes, are often associated with reported improvements in school ethos or discipline, but are not included in this summary, which is limited to interventions that focus directly on behaviour.

Key findings

1. Both targeted interventions and universal approaches have positive overall effects. Schools should consider the appropriate combination of behaviour approaches to reduce overall disruption and provide tailored support where required.

2. There is evidence across a range of different interventions with highest impacts for approaches that have an academic component or focus on role-play and rehearsal.

3. Even within programme types there is a range of impact. If selecting a behaviour intervention, schools should look for programmes or approaches that have been evaluated and shown to have a positive impact where possible.

4. When adopting behaviour interventions – whether targeted or universal – professional development may support staff to ensure high quality delivery and consistency across the school.

How effective is the approach?

The average impact of behaviour interventions is three additional months' progress over the course of a year.

Evidence suggests that, on average, behaviour interventions can produce moderate improvements in academic performance along with a decrease in problematic behaviours. However, estimated benefits vary widely depending on what happens during the intervention.

Approaches such as improving teachers' behaviour management and pupils' cognitive and social skills are both effective, on average. School-level behaviour approaches are often related to improvements in attainment, but there is a lack of evidence to show that the improvements are actually caused by the behaviour interventions, rather than other school interventions happening at the same time.

Behind the average

Closing the disadvantage gap

According to figures from the Department for Education, pupils who receive Free School Meals are more likely to receive a permanent or fixed period exclusion compared to those who do not. In recent data, the suspension rate for pupils eligible for Free School Meals was more than four times higher than for their peers, while permanent exclusion rates were around five times higher.

The most common reason for exclusion is persistent disruptive behaviour. Pupil behaviour will have multiple influences, some of which teachers can directly manage through universal or classroom management approaches. Some pupils will require more specialist support to help manage their self-regulation or social and emotional skills.

How could you implement in your setting?

Behaviour interventions have an impact through increasing the time that pupils have for learning. This might be through reducing low-level disruption that reduces learning time in the classroom or through preventing exclusions that remove pupils from school for periods of time. If interventions take up more classroom time than the disruption they displace, engaged learning time is unlikely to increase. In most schools, a combination of universal and targeted approaches will be most appropriate:

Across all approaches it is crucial to maintain high expectations for pupils and to embed a consistent approach across the school. Successful approaches may also include social and emotional learning interventions.

Evidence suggests that programmes delivered over two-to-six months seem to produce more long-lasting results. Whole-school strategies are usually longer to embed than individually tailored or single-classroom strategies.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School's Guide to Implementation](#).

What does it cost

The costs of behaviour interventions vary widely and overall they are estimated to range between very low to moderate. The costs to schools to deliver whole school strategies are largely based on staff time and training. More intensive, targeted interventions are likely to incur higher staffing and training costs.

Behavioural interventions can require a large amount of staff time, compared with other approaches. Targeted or one-to-one approaches, delivered by trained school staff or specialists, will require additional staff time compared to universal approaches. Overall, effective approaches can promote better engagement with teaching and learning by reducing challenging behaviour and improving pupil engagement.

Alongside time and cost, school leaders should reflect on the impact of whole school behaviour policies and support their staff to maintain a consistent approach. When adopting new approaches, school leaders should consider programmes with a track record of effectiveness. Improving classroom management may involve

intensive training where teachers reflect on their practice, implement new strategies, and review their progress over time.

How secure is the evidence?

The security of the evidence around behaviour interventions is rated as moderate. 83 studies were identified that met the inclusion criteria for the Toolkit. Overall, the topic lost one additional padlock because a large percentage of the studies were not independently evaluated. Evaluations conducted by organisations connected with the approach – for example, commercial providers, typically have larger impacts, which may influence the overall impact of the strand.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

A collaborative (or cooperative) learning approach involves pupils working together on activities or learning tasks in a group small enough to ensure that everyone participates. Pupils in the group may work on separate tasks contributing to a common overall outcome, or work together on a shared task. This is distinct from unstructured group work.

Some collaborative learning approaches put pairs, groups or teams of mixed attainment to work in competition with each other in order to drive more effective collaboration. There is a very wide range of approaches to collaborative and cooperative learning involving many different kinds of organisation and tasks. [Peer tutoring](#) can also be considered as a type of collaborative learning, but is reviewed as a separate topic in the Toolkit.

Key findings

1. Collaborative learning approaches have a positive impact, on average, and may be a cost-effective approach for raising attainment

2. Pupils need support and practice to work together; it does not happen automatically. Professional development can support the effective management of collaborative learning activities.
3. Tasks and activities need to be designed carefully so that working together is effective and efficient, otherwise some pupils may struggle to participate or try to work on their own. It is important to ensure that all pupils talk and articulate their thinking in collaborative tasks to ensure they benefit fully.
4. Competition between groups can be used to support pupils in working together more effectively. However, overemphasis on competition can cause learners to focus on winning rather than succeeding in their learning.
5. The most promising collaborative learning approaches tend to have group sizes between 3 and 5 pupils and have a shared outcome or goal.

How effective is the approach?

The impact of collaborative approaches on learning is consistently positive, with pupils making an additional 5 months' progress, on average, over the course of an academic year. However, the size of impact varies, so it is important to get the detail right.

Collaborative learning can describe a large variety of approaches, but effective collaborative learning requires much more than just sitting pupils together and asking them to work in pairs or group; structured approaches with well-designed tasks lead to the greatest learning gains.

There is some evidence that collaboration can be supported with competition between groups, but this is not always necessary, and can lead to learners focusing on the competition rather than the learning it aims to support. Most of the positive approaches include the promotion of talk and interaction between learners.

The evidence indicates that groups of 3 – 5 is most effective for collaborative learning approaches – there are smaller positive impacts for both paired work and collaborative learning activities with more than 5 pupils in a group. There is also some evidence that collaborative learning approaches are particularly promising when used to teach science.

Behind the average

Closing the disadvantage gap

There is limited evidence on differential impact for pupils from disadvantaged backgrounds. There is some evidence that collaborative learning approaches may benefit those with low prior attainment by providing opportunities for pupils to work with peers to articulate their thinking, share knowledge and skills and address misconceptions through peer support and discussion.

It is crucial that support is provided through well-structured and carefully designed learning activities to ensure that lower-attaining pupils are involved, challenged and learn successfully. If collaborative learning approaches just involve high attaining pupils solving problems with no input from their peers – this is likely to widen existing gaps in attainment.

Applications and approaches

Collaborative learning with joint outcomes

+7

Collaborative learning with individual outcomes

+4

How could you implement in your setting?

There are many theories about how collaborative learning might benefit pupil outcomes. Through collaboration, pupils may develop explanation, demonstration, problem-solving, and metacognitive skills, or pupils may benefit from sharing the load of challenging tasks. It is important that schools ensure that within collaborative learning:

- all pupils, particularly pupils with low prior attainment, are supported to fully participate.
- the make-up of pairings and groups is carefully considered.
- teachers promote good practice in collaboration – for example modelling high quality discussions so that collaborative activities are productive.
- teachers carefully monitor collaborative activities and support pupils that are struggling or not contributing.

There is a broad range of approaches to collaborative or cooperative learning involving different kinds of organisation and tasks across the curriculum. Not all of the specific approaches to collaborative learning adopted by schools have been evaluated, so it is important to evaluate any new initiative in this area.

Professional development is likely to be required to maximise the effectiveness of approaches and monitor the impact of different approaches in the classroom.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School's Guide to Implementation](#).

What does it cost

The average cost of collaborative learning is expected to be very low with the cost to schools largely based on teacher training and resources. As a classroom-based approach, implementing collaborative learning will also require a small amount of staff time for planning and monitoring, compared with other approaches.

Alongside time and cost, school leaders should consider how to maximise the effectiveness of collaborative learning through teacher professional development to support the use of well-designed tasks and should carefully monitor the impact of approaches on lower-attaining pupils.

How secure is the evidence?

The security of the evidence around collaborative learning interventions is rated as low. 212 studies were identified that meet the inclusion criteria of the Toolkit. The topic lost three padlocks because:

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Extending school time involves increasing allocated learning time in school for some or all pupils. School time can be extended in various ways, including:

- Targeted before- and after-school programmes (including small group or one to one tuition);
- Lengthening the school day;
- Providing schooling over the weekend; and
- Increasing the number of days in the school year or providing year-round schooling.

While enrichment activities may be beneficial for their own sake, the studies in this strand focus on extending school time with learning activities. The academic activities in this evidence base are sometimes combined with other non-academic activities.

Extending school time might lead to improved attainment through additional learning hours providing pupils with more exposure to teaching, more time to engage with content, and generally more learning.

Other approaches to increasing learning time are included in other strands of the Toolkit, such as homework, and

summer schools. Breakfast clubs or one-off extended learning opportunities (such as school trips) are not included in this strand.

Key findings

1. Programmes that extend school time have a positive impact on average but are expensive and may not be cost-effective for schools to implement. Schools will also need to consider the workload and wellbeing of their staff.
2. Planning to get the most from any extra time is important. It should meet pupils' needs and build on their capabilities. Where pupils attend voluntarily, it can be helpful to monitor attendance to ensure that the pupils who need additional support are benefitting.
3. When rolling out any new approach, promoting buy-in from pupils, parents and staff is a key consideration. This is especially relevant for extending school time using before- and after-school programmes, where attracting and retaining pupils can be challenging.
4. When extending school time, it is important to acknowledge that scheduled learning time does not always translate into actual learning time. What seems to matter most is how extra time is used – not how much is added. Evidence does not suggest that simply increasing school hours leads to better attainment.
5. Optional additional time runs the risk of not engaging all groups. There is some evidence that combining academic learning with other types of activity, such as physical, cultural/artistic, vocational or social activities, can increase outcomes for pupils from disadvantaged backgrounds.

How effective is the approach?

The average progress figure of 3 months over a school year suggests this approach can be effective. However, while the impact on academic attainment is, on average, positive, the cost of extending school time might mean

that it is not a cost-effective approach to implement at the school level without additional funding. The impact is also slightly lower when school time is extended in secondary school.

Evidence suggests that there is not a clear relationship between the amount of additional time and its impact on attainment. Schools should consider how to ensure additional time benefits learning. There are a small number of examples where adding large amounts of additional time decreases results for pupils who are socioeconomically disadvantaged or struggling academically. It may be that extreme increases in time have diminishing effects if the engagement of pupils is reduced.

Some wider research evidence suggests that participation in enrichment activities is associated with progression to higher education and being in employment.

Behind the average

Closing the disadvantage gap

To increase the likelihood of additional school time benefitting disadvantaged pupils, school leaders should consider how to secure engagement and attendance among those from disadvantaged backgrounds. It is possible that if targeted tuition or enrichment activities are offered universally, those who could benefit the most would be the least likely to participate or engage. Adopting a more targeted approach also has its challenges though, as selected pupils may feel singled out and stigmatised.

Sometimes, additional activities delivered alongside academic intervention seem to have positive results on outcomes.

Examples from the evidence base of enrichment used alongside academic activities include:

- Physical activities: sports, fitness, dance;
- Cultural or artistic activities: music, arts and crafts, drama;
- Vocational activities: computer skills, careers, entrepreneurship; and
- Social activities: games, groupwork.

How could you implement in your setting?

When extending school time, it is important to acknowledge that scheduled learning time does not always translate into actual learning time. In fact, research evidence also suggests that there is not a relationship between the amount of time added and the resulting impact on attainment. The quality of the extra time appears more important than the quantity of extra time.

To help make the best use of extra time, schools should:

Extending school time is likely to require a significant reconfiguration of working patterns for staff, especially if this involves an altered school calendar. It is important school leaders are clear on the purpose of introducing additional learning time.

Approaches to extending school time are likely to be spread over an academic year. Some schools may also target additional support at specific classes or pupils during particular school terms or times of the year.

If no additional staff are recruited to cover extended school time, then implementing this approach will require significantly more

staff time than other Toolkit strands. Staff already working at a school have good knowledge of their curriculum and their pupils, but this should be balanced with the impact on existing staff. When introducing new approaches, schools should consider implementation. For more information see [A School's Guide to Implementation](#).

What does it cost

Overall, costs are estimated as moderate. However, without additional funding, programmes that extend school time can be expensive and may not be cost-effective for schools to implement. The basic cost of teaching a pupil is about £3120 a year (£16 per day) in primary school and about £4,680 a year (£25 per day) in secondary. Extending the school year by two weeks would therefore require about £160 per pupil per year for primary schools and about £250 per pupil per year for secondary. Estimates suggest after-school clubs cost, on average, £7 per session per pupil. Extrapolating this, one weekly session of extra learning could therefore cost £273 per pupil over the course of a 39-week school year. Extending school time can also involve additional costs other than staffing, such as providing transportation for pupils.

How secure is the evidence?

The security of the research evidence around extending school time is rated as moderate. 92 studies were identified that meet the inclusion criteria of the Toolkit. Overall, the topic lost an additional padlock because a large percentage of the studies are not randomised controlled trials. While other study designs still give important information about effectiveness of approaches, there is a risk that results are influenced by unknown factors that are not part of the intervention.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Feedback is information given to the learner about the learner's performance relative to learning goals or outcomes. It should aim to (and be capable of producing) improvement in students' learning.

Feedback redirects or refocuses the learner's actions to achieve a goal, by aligning effort and activity with an outcome. It can be about the output or outcome of the task the process of the task the student's management of their learning or self-regulation, or about them as individuals (which tends to be the least effective).

This feedback can be verbal or written, or can be given through tests or via digital technology. It can come from a teacher or someone taking a teaching role, or from peers (see [Peer tutoring](#)).

Key findings

1. Providing feedback is a well-evidenced and has a high impact on learning outcomes. Effective feedback tends to focus on the task, subject and self-regulation strategies: it provides specific information on how to improve.
2. Feedback can be effective during, immediately after and some time after learning. Feedback policies should not over specify the frequency of feedback
3. Feedback can come from a variety of sources – studies have shown positive effects of feedback from teachers and peers. Feedback delivered by digital technology also has positive effects (albeit slightly lower than the overall average).
4. Different methods of feedback delivery can be effective and feedback should not be limited exclusively to written marking. Studies of verbal feedback show slightly higher impacts overall (+7 months). Written marking may play one part of an effective feedback strategy – but it is crucial to monitor impacts on staff workload.
5. It is important to give feedback when things are correct – not just when they are incorrect. High-quality feedback may focus on a task, subject, and self-regulation strategies.

How effective is the approach?

Feedback studies tend to show high effects on learning. However, there are a wide range of effects and some studies show that feedback can have negative effects and make things worse.

There are positive impacts from a wide range of feedback approaches – including when feedback is delivered by technology or peers. Impacts are highest when feedback is delivered by teachers. It is particularly important to provide feedback when work is correct, rather than just using it to identify errors.

Many studies of feedback also include other practices. For example, mastery learning approaches combine feedback with additional support for pupils that are falling behind, while approaches like formative assessment also include work to understand specific gaps in learning that need to be addressed and how the teacher wants the pupil to progress.

Feedback has effects across all age groups. Research in schools has focused particularly on its impact on English, mathematics and, to a lesser extent, science.

Embedding formative assessment explicitly can be a key component of laying the foundations for effective feedback. The EEF has trialled 'Embedding Formative Assessment' in English schools and found a positive impact, on average.

Behind the average

Closing the disadvantage gap

There is evidence to suggest that feedback involving metacognitive and self-regulatory approaches may have a greater impact on disadvantaged pupils and lower prior attainers than other pupils. Pupils require clear and actionable feedback to employ metacognitive strategies as they learn, as this information informs their understanding of their specific strengths and areas for improvement, thereby indicating which learning strategies have been effective for them in previously completed work.

Applications and approaches

Written feedback

+5

Oral feedback

+7

How could you implement in your setting?

Feedback may have a positive impact through supporting pupils to focus future learning on areas of weakness, through identifying and explaining misconceptions, through supporting them in taking greater responsibility for their own improvement or through increasing pupils' motivation to improve.

Implementing feedback successfully will require:

- communicate with pupils, teachers, and parents, about practices and expectations that relate to feedback policies,
- assessing pupil understanding, so that you know what needs to be improved,
- consider the 'opportunity cost' associated with different feedback practices,
- ensuring that feedback can be acted upon, for example through including specific information on what a pupil has done successfully or not, assisted with an explanation as to why,
- carefully considering how feedback will be received, including impacts on self-confidence and motivation,
- providing opportunities for pupils to act-upon the feedback after it has been given, evaluating how effective the feedback has been.

Feedback interventions vary in length. Some function as short, targeted approaches that address pupil misconceptions within the space of weeks or even days. Alternatively, others are used as more extended methods of tracking and supporting pupil progress over many months.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School's Guide to Implementation.](#)

What does it cost

The average cost of feedback and feedback interventions is very low. The cost to schools is largely based on training.

Implementing feedback and feedback interventions will also require a moderate and sustained amount of staff time, compared with other approaches.

Alongside time and cost, school leaders should consider how to maximise teacher professional development in supporting them to deliver effective feedback and avoid approaches that increase teacher workload without providing pupils with the necessary information to improve performance.

How secure is the evidence?

The security of the evidence around feedback is rated as high. 155 studies were identified that meet the inclusion criteria of the Toolkit. Overall, the topic lost one padlock because a large percentage of the studies are not randomised controlled trials. While other study designs still give important information about effectiveness of approaches, there is a risk that results are influenced by unknown factors that are not part of the intervention.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Homework refers to tasks given to pupils by their teachers to be completed outside of usual lessons.

Homework activities vary significantly, particularly between younger and older pupils, including but not limited to home reading activities, longer projects or essays and more directed and focused work such as revision for tests.

Our definition also includes activities such as 'homework clubs' where pupils have the opportunity to complete homework in school but outside normal school hours, and 'flipped learning' models, where pupils prepare at home for classroom discussion and application tasks.

Key findings

1. Homework has a positive impact on average (+ 5 months), particularly with pupils in secondary schools.
2. Some pupils may not have a quiet space for home learning – it is important for schools to consider how home learning can be supported (e.g. through providing homework clubs for pupils).

3. Homework that is linked to classroom work tends to be more effective. In particular, studies that included feedback on homework had higher impacts on learning.

4. It is important to make the purpose of homework clear to pupils (e.g. to increase a specific area of knowledge, or to develop fluency in a particular area).

How effective is the approach?

The average impact of homework is positive across both primary and secondary school. There is, however variation behind this average with homework set in primary school having a smaller impact on average (see below).

The quality of the task set appears to be more important than the quantity of work required from the pupil. There is some evidence that the impact of homework diminishes as the amount of time pupils spend on it increases. The studies reviewed with the highest impacts set homework twice a week in a particular subject.

Evidence also suggests that how homework relates to learning during normal school time is important. In the most effective examples homework was an integral part of learning, rather than an add-on. To maximise impact, it also appears to be important that students are provided with high quality feedback on their work (see Feedback).

Behind the average

Closing the disadvantage gap

Pupils eligible for free school meals typically receive additional benefits from homework. However, surveys in England suggest that pupils from disadvantaged backgrounds are less likely to have a quiet working space, are less likely to have access to a device suitable for learning or a stable internet connection and may receive less parental support to complete homework and develop effective learning habits. These difficulties may increase the gap in attainment for disadvantaged pupils.

Homework clubs can help to overcome these barriers by offering pupils the resources and support needed to undertake homework or revision. Broader evidence suggests that homework should not be used as a punishment or penalty for poor performance.

Applications and approaches

Homework in primary schools

+3

Homework in secondary schools

+5

How could you implement in your setting?

Homework has an impact by enabling pupils to undertake independent learning to practice and consolidate skills, conduct in-depth inquiry, prepare for lessons or revise for exams. When implementing homework, the evidence suggests a wide variation in impact. Therefore, schools should consider the ‘active’ ingredients to the approach, which may include:

- Considering the quality of homework over the quantity.
- Using well-designed tasks that are linked to classroom learning.
- Clearly setting out the aims of homework to pupils.
- Understanding and addressing any barriers to completion, such as access to a learning device or resources.
- Explicitly teaching independent learning strategies.
- Providing high-quality feedback to improve pupil learning.
- Monitoring the impact homework on pupil engagement, progress and attainment.

Teachers should seek to understand any barriers to completing homework – for example, a lack of access to a quiet space or learning materials – and aim to avoid approaches that use homework as a penalty for poor performance. When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School’s Guide to Implementation](#).

What does it cost

The average cost of homework is expected to be very low with the cost to schools largely based on teacher training and resources. Implementing homework will also require a small amount of staff time for planning and feedback.

Alongside time and cost, school leaders should consider how to maximise the effectiveness of homework through teacher professional development to promote the use of well-designed tasks to complement learning in the classroom and high-quality feedback to improve pupil learning. Schools should monitor the impact of different approaches to homework – such as the frequency, purpose and variety of tasks – on pupil engagement and attainment.

How secure is the evidence?

The security of the evidence around homework is rated as low. 43 studies were identified that meet the inclusion criteria of the Toolkit. The topic lost additional padlocks because:

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Individualised instruction involves providing different tasks for each learner and support at the individual level. It is based on the idea that all learners have different needs, and that therefore an approach that is personally tailored – particularly in terms of the activities that pupils undertake and the pace at which they progress through the curriculum – will be more effective.

Various models of individualised instruction have been researched over the years in education, particularly in subjects like mathematics where pupils can have individual sets of activities which they complete, often largely independently. More recently, digital technologies have been employed to facilitate individual activities and feedback.

Key findings

1. Individualised instruction can be an effective approach to increasing pupil attainment. It can, however, be a challenging approach to implement given the increased requirements on the teacher to organise and monitor individual activities.

2. Studies of Individualised instruction with older pupils of secondary age tend to show higher effects. It may be that the impact is increased when pupils are more skilled at managing their own learning.

3. There is evidence that digital technology can be used effectively to provide individualised instruction. Many of these studies use digital technology alongside small group tuition, with teachers providing targeted instruction to the pupils that are not engaging with the technology.

4. Small group learning might be another promising approach to meeting differing learner needs without reducing the total amount of teaching time that pupils receive.

How effective is the approach?

On average, individualised instruction approaches have an impact of 4 months' additional progress.

Behind this average, there is a large amount of variation. Some of this may be explained by the challenges of implementing the approach effectively, without diminishing engaged learning time. For classroom-based approaches, it appears that the role of the teacher may become more managerial, with the increased need for organising and monitoring learning activities leaving less time for high quality pedagogical interaction. Because of this, individualised instruction may be better used as a supplement to usual class teaching, rather than a replacement.

Some recent studies have used digital technology with diagnostic assessment and feedback to individualise instruction, and positive impacts on average. For example, technology may enable more

immediate feedback on the individualised tasks (for more detail on the impact of feedback see [here](#)).

A small number of studies have examined including peer feedback as part of individualised instruction. The results in these studies are positive, on average.

Behind the average

Closing the disadvantage gap

Disadvantaged pupils may be more likely to be affected by higher school absence rates and lower attainment, on average, when compared to their more advantaged peers. This means disadvantaged pupils may be more likely to progress through school with gaps in their understanding, which will affect how quickly and easily they can acquire and connect new learning.

For pupils identified as having low prior attainment or at risk of falling behind, individualised instruction may allow the teacher to provide activities that are closely matched to a pupil's attainment. Provided they have the skills to manage their learning independently, this can support pupils to consolidate their learning and practice skills or develop mastery before progressing to the next stage of the curriculum. More targeted assessment and feedback may also support pupils to address misconceptions or overcome specific barriers to learning.

How could you implement in your setting?

Individualised instruction aims to improve outcomes through providing targeted support to learners. In order to ensure that the approach is effective, schools should consider how they will provide:

In some studies, these elements have been provided through digital technology – for example, through intelligent tutoring systems that provide responsive feedback and assessment.

Individualised instruction interventions can be delivered through a range of models including independent learning, classroom-based activities supported by a teacher or teaching assistant and the use of digital technologies which have been developed to support individual activities with assessment and feedback.

School leaders should consider how best to monitor approaches to ensure individualised instruction is implemented effectively.

Ensuring individualised instruction activities are used to supplement (and not replace) high-quality teacher interaction is important. School leaders may also consider small group approaches to provide learners with effective practice, monitored and supported by a teaching assistant.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School's Guide to Implementation](#).

What does it cost

The average cost of individualised instruction is very low. The costs to schools are largely based on teacher professional development, training and the cost associated with digital resources and software for individualised approaches.

Adopting individualised instruction will also require a small amount of additional staff time compared with other approaches as interventions are largely delivered during lesson time.

How secure is the evidence?

The security of the evidence around Individualised Instruction is rated as limited. 198 studies were identified that meet the inclusion criteria for the Toolkit. Overall, the topic lost three additional padlocks because:

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

The idea underpinning learning styles is that individuals all have a particular approach to or style of learning. The theory is that learning will therefore be more effective or more efficient if pupils are taught using the specific style or approach that has been identified as their learning style. For example, pupils categorised as having a 'listening' learning style, could be taught more through storytelling and discussion and less through traditional written exercises.

Key findings

1. The number of high-quality studies of learning styles are extremely low. As a result no impact is displayed. Schools implementing approaches with very limited evidence should carefully consider how they would monitor impact and mitigate against the risk of negative effects.
2. Learners are very unlikely to have a single learning style, so restricting pupils to activities matched to their reported preferences may damage their progress. This is especially true for younger learners in primary schools whose preferences and approaches to learning are still very flexible.

3. Labelling students as particular kinds of learners is likely to undermine their belief that they can succeed through effort and to provide an excuse for failure.
4. It appears to be more promising to focus on other aspects of motivation and self-regulation to engage pupils in learning activities.
5. Teachers should aim to support pupils to take responsibility for success in their learning and develop their own successful strategies and approaches.

How effective is the approach?

The lack of studies identified that tested learning styles approaches rigorously, as found through a systematic search, mean that there is not enough security to communicate a month's progress figure.

There is very limited wider evidence for any consistent set of learning 'styles' that can be used reliably to identify genuine differences in the learning needs of young people. Evidence also suggests that it is unhelpful to assign learners to groups or categories based on a supposed learning style. It is particularly important not to label primary age pupils, or for them to be led to believe that any lack of success is due to their learning styles.

Individual learning preferences do change in different situations and over time, and there is some evidence that cognitive preference and task type may be connected (for example, visualisation is particularly valuable for some areas of mathematics). However, studies where teaching activities are targeted towards particular learners based on an identified learning 'style' have not convincingly shown any major benefit, particularly for low attaining pupils. Impacts recorded are generally low or negative.

Closing the disadvantage gap

Studies where teaching activities are targeted towards particular learners based on an identified learning 'style' have not indicated an impact on pupil attainment and therefore, grouping pupils on the basis of learning styles is unlikely to be a successful strategy for closing the disadvantage attainment gap.

Adaptive teaching approaches are unlikely to be valuable if teachers set lower expectations for particular pupils. It is important not to label younger learners or to attribute poor performance to their 'learning style' as this may negatively impact pupil motivation and self-efficacy. This poses a particular risk for pupils from disadvantaged backgrounds who are, on average, more likely to have lower prior attainment.

How could you implement in your setting?

There is very limited evidence for any consistent set of learning 'styles' that can be used reliably to identify the learning needs of young people. Instead, teachers hoping to target learning effectively might consider other teaching and learning practices including:

As a classroom-based approach, activities are typically delivered by teachers or TAs.

What does it cost

The costs are estimated as very low, usually involving preparation of a greater range and variety of teaching and learning materials. Some of the available tests of learning styles require purchase and it is important to be aware of the lack of validity and reliability of these tests given the lack of evidence for the existence of learning styles noted above.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School's Guide to Implementation](#).

How secure is the evidence?

The lack of identified studies that tested learning styles approaches rigorously mean that the security of the evidence around Learning Styles is rated as extremely low. For topics with extremely low evidence, a month's progress figure is not displayed. No studies were identified that met the pre-specified inclusion criteria.

What is it?

Mastery learning was originally developed in the 1960s. According to the early definition of mastery learning, learning outcomes are kept constant but the time needed for pupils to become proficient or competent at these objectives is varied.

Subject matter is broken into blocks or units with predetermined objectives and specified outcomes. Learners must demonstrate mastery on unit tests, typically 80%, before moving on to new material. Any pupils who do not achieve mastery are provided with extra support through a range of teaching strategies such as more intensive teaching, tutoring, peer-assisted learning, small group discussions, or additional homework. Learners continue the cycle of studying and testing until the mastery criteria are met.

More recent mastery approaches do not always have all these characteristics of mastery learning. Some approaches without a threshold typically involve the class moving on to new material when the teacher decides that the majority of pupils have mastered the unit. Curriculum time is varied according to the progress of the class. In other approaches, pupils are required to demonstrate mastery on a test to progress to new material, but there is not a specified threshold of at least 80%.

Mastery Learning should be distinguished from a related approach sometimes known as “teaching for mastery”. This term is often used to describe the approach to maths teaching found in high-performing places in East Asia, such as Shanghai and Singapore. Like “mastery learning”, “teaching for mastery” aims to support all pupils to achieve deep understanding and competence in the relevant topic. However, “teaching for mastery” is characterised by teacher-led, whole-class teaching; common lesson content for all pupils; and use of manipulatives and representations. Although

some aspects of “teaching for mastery” are informed by research, relatively few interventions of this nature have been evaluated for impact. Most of the studies in this strand, should be distinguished from this related approach.

Key findings

1. Mastery learning is a cost-effective approach, on average, but is challenging to implement effectively. Schools should plan for changes and assess whether the approach is successful within their context.
2. A high level of success should be required before pupils move on to new content – it is crucial to monitor and communicate pupil progress and to provide additional support for pupils that take longer to reach the required level of knowledge.
3. Mastery learning approaches are often associated with direct instruction, but many of the high impact studies identified included elements of collaborative learning.
4. There is large variation behind the average impact – mastery learning approaches have consistently positive impacts, but effects are higher for primary school pupils and in mathematics.

How effective is the approach?

The impact of mastery learning approaches is an additional five months progress, on average, over the course of a year.

There is a lot of variation behind this average. It seems to be important that a high bar is set for achievement of ‘mastery’ (usually 80% to 90% on the relevant test). By contrast, the approach appears to be much less effective when pupils work at their own pace (see also [Individualised instruction](#)).

Mastery learning also appears to be particularly effective when pupils are given opportunities to work in groups or teams and take responsibility for supporting each other’s progress (see also [Collaborative learning](#) and [Peer tutoring](#)).

Behind the average

Closing the disadvantage gap

Mastery learning approaches aim to ensure that all pupils have mastered key concepts before moving on to the next topic – in contrast with traditional teaching methods in which pupils may be left behind, with gaps of misunderstanding widening. Mastery learning approaches could address these challenges by giving additional time and support to pupils who may have missed learning, or take longer to master new knowledge and skills.

In order for mastery approaches to be effective for pupils with gaps in understanding, it is crucial that additional support is provided. Approaches that simply build upon foundational knowledge without targeting support for pupils that fall behind are unlikely to narrow disadvantage gaps.

How could you implement in your setting?

Mastery learning works through designing units of work so that each task has a clear learning outcome, which pupils must master prior to moving on to the next task. Core components of the mastery approach that schools should be careful to implement include:

Mastery learning interventions are typically delivered over the course of an academic year, as choosing to take longer on topic or scheme of work requires flexibility in the planning and teaching of curriculum content.

Some schools may decide that certain topics are more suited to a mastery approach than others, and therefore the delivery time could be as short as half a term.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School’s Guide to Implementation](#).

What does it cost

Overall, the median costs of implementing mastery learning approaches are estimated as very low. The costs associated with mastery learning approaches mostly arise from professional development training for teaching staff, which is most commonly a start-up cost for introducing the new approach.

Whilst the average cost estimate for mastery learning is very low, the range in costs of professional development training, and the option to pay for ongoing training and additional staff to provide greater timetable flexibility, mean that costs can range from very low to moderate.

Implementing mastery learning will also require a moderate amount of staff time, compared with other approaches. School leaders should be aware of the extra staff time required and think carefully about other activities they might need to cut back on to provide this additional support.

Alongside time and cost, school leaders should consider how to maximise support for struggling learners and avoid some pupils getting bored or frustrated whilst they wait for others to master content.

How secure is the evidence?

The security of the evidence around mastery learning is rated as low. 80 studies were identified that meet the inclusion criteria of the Toolkit. Overall, the topic lost two additional padlocks because:

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Mentoring in education involves pairing young people with an older peer or adult volunteer, who acts as a positive role model. In general, mentoring aims to build confidence and relationships, to develop resilience and character, or raise aspirations, rather than to develop specific academic skills or knowledge.

Mentors typically build relationships with young people by meeting with them one to one for about an hour a week over a sustained period, either during school, at the end of the school day, or at weekends. In some approaches mentors may meet with their mentees in small groups.

Activities vary between different mentoring programmes. While some mentoring programmes include academic support with homework or other school tasks, approaches focused primarily on direct academic support (sometimes referred to as “academic mentoring”) are not covered in this strand. See [one to one tuition](#) and [peer tutoring](#).

Mentoring has increasingly been offered to young people who are deemed to be hard to reach or at risk of educational failure or exclusion.

Key findings

1. The impact of mentoring varies but, on average, it is likely to have a small positive impact on attainment (+2 months' additional progress).

2. Positive effects on attainment tend not to be sustained once the mentoring stops, so care must be taken to ensure that benefits are not lost. It is important to consider how you can support pupils who have benefitted from mentoring to retain positive changes in their confidence and behaviour.

3. Both community-based and school-based approaches can be successful, though most of the studies included in this strand took place in schools

4. Mentor drop-out can have detrimental effects on mentees. It is important to consider how to support mentors.

How effective is the approach?

On average, mentoring appears to have a small positive impact on academic outcomes. The impacts of individual programmes vary. Some studies have found more positive impacts for pupils from disadvantaged backgrounds, and for non-academic outcomes such as attitudes to school, attendance and behaviour.

There are risks associated with unsuccessful mentor pairings, which may have a detrimental effect on the mentee, and some studies report negative overall impacts.

Programmes which have a clear structure and expectations, provide training and support for mentors, and recruit mentors who are volunteers, are associated with more successful outcomes.

There is no evidence that approaches with a single focus on improving academic attainment or performance are more effective.

Programmes with multiple objectives can be equally or more effective.

Behind the average

Closing the disadvantage gap

While mentoring is not generally as effective in raising attainment outcomes as small group or one to one tuition, it is possible to target the approach to pupils from disadvantaged backgrounds and those with particular needs.

Some evidence suggests that some pupils from disadvantaged backgrounds show low engagement with or have low expectations of schooling. Mentoring interventions may be more beneficial for these pupils, as the development of trusting relationships with an adult or older peer can provide a different source of support.

Mentors dropping out of programmes can result in detrimental effects for pupils. For pupils who may have already experienced disillusion at their perceived lack of support from teachers and school, this risk may be intensified.

Therefore, additional care should be given to the recruitment of reliable mentors when interventions are being used to support disadvantaged pupils.

How could you implement in your setting?

Mentoring requires close interaction between an adult or older peer and one or a small group of pupils. Conversations between mentors and mentees may address but would not be limited to: attitudes to school; specific academic skills or knowledge; self-perception and belief, particularly in relation to school-work; aspirations for future studies and career options. It is important to consider what support mentors might require to effectively deliver mentoring. Careful consideration should be given to the type of training mentors need to support them build successful mentoring relationships.

Mentoring interactions normally occur one to one between mentor and mentee – although mentors can mentor multiple pupils. Some mentoring approaches also include small group interactions with groups of two to five students, and extremely rarely with groups of more than five students.

Mentoring interventions are typically delivered over an extended period of time (often at least the length of a school year) in order to allow mentors and mentees to develop more lasting and trusting relationships. Frequent regular meetings of once a week or more tend to be more beneficial.

When introducing new approaches, schools should consider implementation. For more information see [A School's Guide to Implementation](#).

What does it cost

The average cost of a mentoring intervention is moderate. The cost to schools is largely based on mentor training, salary costs (for non-volunteer mentors) and resources. Some programmes also include continuous training and support for mentors which may increase costs.

Implementing mentoring interventions will also require a moderate and sustained amount of staff time, compared with other approaches.

Alongside time and cost, school leaders should consider how to maximise the recruitment of effective and reliable mentors that are well matched to mentees. Consideration should also be given to how any gains made in pupil confidence, resilience or aspiration are to be maintained after the intended period of mentoring, as studies show these changes can be difficult to sustain.

How secure is the evidence?

The security of the evidence around mentoring is rated as moderate. 64 studies were identified that meet the inclusion criteria of the Toolkit.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Metacognition and self-regulation approaches support pupils to think about their own learning more explicitly, often by teaching them specific strategies for planning, monitoring and evaluating their learning.

- Metacognition is the learner's ability to be aware of, reflect on, and direct their thinking.
- Self-regulated learners apply metacognitive strategies to their learning. They demonstrate self-regulation by managing their motivation, thoughts and behaviour to set goals, monitor working, reflect and review progress.

Studies focus on equipping pupils with metacognitive strategies. This promotes learner independence. Teachers have an important role to play in the explicit teaching of metacognition and self-regulation strategies. Teachers can also model metacognition and provide scaffolding to support pupils to become better self-regulated learners.

Other toolkit strands such as [Oral Language Interventions](#), Reading [Comprehension Strategies](#) and [Collaborative Learning Approaches](#)

are closely linked to Metacognition and Self-Regulation. These approaches feature studies that use strategies and practices similar to those in Metacognition and Self-regulation.

Metacognition and self-regulated learning is a crucial element of teaching but can be a complex topic; the EEF have produced further guidance which you may find useful, '[Metacognition and Self-Regulated Learning: Guidance Report](#)'.

Key findings

1. The average impact of metacognition and self-regulation approaches is high (+8 months additional progress) however, it can be difficult to realise this impact in practice. Teachers should be supported to both explicitly teach metacognitive strategies and to promote metacognitive skills in lessons.
2. The research evidence indicates that explicitly teaching strategies to help pupils to plan, monitor and evaluate specific aspects of learning can be effective. Research suggests these should be taught and applied to the usual curriculum content, rather than taught discretely through 'thinking skills' lessons.
3. Teachers can promote metacognition by modelling their thinking aloud. For example, teachers might explain their thinking when planning an exam response or breaking down a mathematical problem, alongside promoting and planning for metacognitive talk related to lesson objectives.
4. In effective studies, teachers often model the explicit use of a metacognitive strategy during a task. For example, they might co-construct a planning checklist and explain how this supports learning, or verbalise self-questioning to evaluate progress.
5. The research evidence suggests that promoting reasoning, argumentation and debate helps to develop metacognition and self-regulation. This is because it encourages pupils to reflect on their learning, the learning of others, and draw connections between topics.

How effective is the approach?

The average impact of metacognition and self-regulation strategies is an additional eight months' progress over the course of a year.

In studies that are effective, metacognition and self-regulation strategies tend to be applied to challenging tasks rooted in the usual curriculum content.

Metacognition and self-regulation strategies can be effective when taught in collaborative groups so that learners can support each other and make their thinking explicit through structured discussion. However, research suggests that this collaboration will need to be carefully structured to ensure it is effective.

Teachers can use a variety of resources to help pupils develop connections between concepts, understand sequential steps, and make criteria for success explicit. For example, teachers might use reflection questions, knowledge organisers, worked examples or success criteria. This supports pupils to plan, monitor and reflect on their learning.

Improved outcomes have been identified across literacy, maths and science. The approach is effective for both primary and secondary schools.

Behind the average

Closing the disadvantage gap

Some studies suggest disadvantaged pupils are less likely to use metacognitive and self-regulation strategies unless explicitly taught to do so, but broader evidence on this link is limited.

The research evidence suggests that studies with higher numbers of disadvantaged pupils had a high impact. This is a promising area of practice to work towards closing the disadvantage gap. Improved metacognition and self-regulation skills have the potential to promote learning across the curriculum and beyond.

How could you implement in your setting?

Metacognition and self-regulation strategies work through learners planning, monitoring and evaluating their learning strategies. Effective teaching practices include:

Metacognition and self-regulation strategies appear more effective when embedded in a school's curriculum and a specific subject lesson. For example, you may teach the use of self-questioning by applying it to your school curricula such as literacy, history and science.

Research evidence suggests that the effectiveness of metacognition and self-regulation strategies is influenced by teachers' understanding of how to develop pupils' metacognitive knowledge. There is some evidence to suggest that teachers should understand the theory behind metacognition and self-regulation; schools should focus on developing a shared understanding of this approach.

More information, tools and resources on Metacognition and Self-regulation can be found in the Guidance Report ['Metacognition and Self-regulated Learning'](#).

There are a range of EEF/ RSN resources and exemplification around this topic that you may find useful. For example:

Whilst this strand includes evidence from children in the Early Years, additional EEF resources offer further support for this age group:

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School’s Guide to Implementation](#). The EEF offers further support for school in designing and implementing effective professional development, ‘[Effective Professional Development](#)’.

What does it cost

Overall, the median costs of implementing metacognition and self-regulation strategies are estimated as very low. The costs associated with metacognition and self-regulation mostly arise from professional development for staff.

Whilst the median cost estimate for metacognition and self-regulation strategies is very low, the range in cost of professional development training, and the option to purchase additional materials and provide ongoing training and support, means that costs can range from very low to low.

This cost estimate assumes that schools are already paying for staff salaries, materials and equipment for teaching, and facilities to host lessons. These are all pre-requisite costs of implementing metacognition and self-regulation strategies, without which the cost is likely to be higher.

Implementing metacognition and self-regulation strategies will also require staff time as staff need to develop their own understanding of metacognition and self-regulation processes to model effective use of these strategies and skills to pupils.

How secure is the evidence?

The security of the evidence around metacognition and self-regulation strategies is rated as high. 355 studies were identified. Overall, the topic lost a padlock because a large percentage of the studies were not independently evaluated. Evaluations conducted by organisations connected with the approach – for example, commercial providers, typically have larger impacts, which may influence the overall impact of the strand.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

One to one tuition involves a teacher, teaching assistant or other adult giving a pupil intensive individual support. It may happen outside of normal lessons as additional teaching – for example as part of extending school time or a summer school – or as a replacement for other lessons.

Key findings

1. On average, one to one tuition is very effective at improving pupil outcomes. One to one tuition might be an effective strategy for providing targeted support for pupils that are identified as having low prior attainment or are struggling in particular areas.
2. Tuition is more likely to make an impact if it is additional to and explicitly linked with normal lessons.
3. One to one tuition can be expensive to deliver, particularly when delivered by teachers. Approaches that either deliver instruction through teaching assistants or in small groups rather than one to one have smaller positive effects, on average, but may be a cost-effective solution to providing targeted support.
4. For one to one tuition led by teaching assistants, interventions are likely to be particularly beneficial when the teaching assistants are experienced, well-trained and supported – for example, delivering a structured intervention.

How effective is the approach?

Evidence indicates that one to one tuition can be effective, providing approximately five additional months' progress on average.

Short, regular sessions (about 30 minutes, three to five times a week) over a set period of time (up to ten weeks) appear to result in optimum impact. Evidence also suggests tuition should be additional to, but explicitly linked with, normal teaching, and that teachers should monitor progress to ensure the tutoring is beneficial. Studies comparing one to one with small group tuition show mixed results. In some cases one to one tuition has led to greater improvement, while in others tuition in groups of two or three has been equally or even more effective. The variability in findings may suggest it is the particular type or quality of teaching enabled by very small groups that is important, rather than the precise size of the group.

Programmes involving teaching assistants or volunteers can have a valuable impact, but may be less effective than those using experienced and specifically trained teachers. Where tuition is delivered by volunteers or teaching assistants there is evidence that training and the use of a structured programme is advisable.

Behind the average

Closing the disadvantage gap

Studies in England have shown that pupils eligible for free school meals typically receive additional benefits from one to one tuition. Low attaining pupils are particularly likely to benefit.

One to one tuition approaches can enable pupils to make effective progress by providing intensive, targeted academic support to those identified as having low prior attainment or at risk of falling behind. The approach allows the teacher or tutor to focus exclusively on the needs of the learner and provide teaching that is closely matched to each pupil's understanding. One to one tuition offers greater levels of interaction and feedback compared to whole class teaching which can support pupils to spend more time on new or unfamiliar, overcome barriers to learning and increase their progress through the curriculum.

How could you implement in your setting?

One to one has an impact by providing additional support that is targeted at a pupil's needs. Reducing the ratio of pupils to teacher allows for closer interaction between educators and pupils. When adopting one to one tuition, schools should consider how to ensure that these active ingredients have a positive impact through:

One to one tuition may be delivered by teachers, trained teaching assistants, academic mentors or tutors. Interventions are typically delivered over an extended period, often over the course of several weeks or a term.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School's Guide to Implementation](#).

What does it cost

The average cost of one to one tuition is moderate. The costs to schools are largely based on additional salary costs and learning resources, the majority of which are recurring costs. Through the National Tutoring Programme Year 1 (2020–21), schools could purchase subsidised in person or online 1:1 tuition in 15hr blocks, for an average cost of £167 – £180 per pupil. Costs are lower for online delivery compared to in person tuition and are higher when provided by qualified or specialist teachers.

When delivering teacher or TA-led Small group tuition, implementation is likely to require a large amount of staff time compared with whole class approaches. Given the lower costs, small group tuition may be a sensible approach to trial before considering one to one tuition. See [small group tuition](#).

Alongside time and cost, school leaders should consider using providers with a track record of effectiveness. To increase the impact of school-led one to one tuition, school leaders might consider professional development for teachers, TAs, and tutors to support high-quality teaching in areas such as formative assessment, curriculum knowledge, instruction and feedback, which will build capacity in school.

How secure is the evidence?

The security of the evidence around one to one tuition is rated as moderate. 123 studies were identified that meet the inclusion criteria for the Toolkit. The topic lost two additional padlocks because:

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Oral language interventions are about making the most of spoken language and verbal interaction for learning.

‘Oral language’ refers to speaking and listening. ‘Intervention’ means doing something new or making a change or refinement to usual practice. Oral language interventions can therefore include approaches that emphasise speaking and/or listening in a whole class setting. They can also include targeted and small-group approaches.

Oral language approaches might include:

- targeted reading aloud and book discussion with young children;
- explicitly extending pupils' spoken vocabulary;
- the use of structured questioning to develop reading comprehension; and
- the use of purposeful, curriculum-focused, dialogue and interaction.

Formal public speaking skills and presentational talk are not explicitly the focus of this research evidence.

What ties the studies in this strand together is the teaching approach, rather than the aim of the intervention. Even though all these interventions make use of oral language, they do not necessarily target pupils' oral language skills as their primary outcome.



In oral language interventions, the adult might be:

- scaffolding pupils' speaking and listening;
- modelling high-quality speaking and listening;
- modelling metacognitive strategies for reading comprehension;
- using dialogic approaches;
- providing feedback; and
- teaching pupils how to use an oral language strategy for learning.



The pupils might be using oral language with the purpose of:

- learning reading or writing skills;
- developing their vocabulary;
- developing their ideas;
- making connections with prior knowledge;
- developing their conversation skills, or other speaking and listening skills; and
- developing their metacognitive awareness.

Other approaches in this Toolkit also involve oral language, such as Metacognition (which makes talk about learning explicit in classrooms), Collaborative learning approaches (which promote pupils' interaction in groups), and Feedback (when delivered verbally).

Key findings

1. On average, oral language approaches have a high impact on pupil outcomes (+6 months' additional progress).

2. Interaction for learning is a key aspect of these interventions.

Approaches often involve spoken interaction between the practitioner and pupils and between peers.

3. The evidence is strongest on interventions that are integrated with the existing curriculum. It is important to consider how any changes or refinements to practice are integrated with the curriculum.

4. Oral language interventions can be an effective approach for disadvantaged pupils. In the research evidence, the higher the proportion of disadvantaged pupils taking part, the more effective the intervention seemed to be.

How effective is the approach?

The average impact of +6 months' progress is based mainly on pupils' scores in tests of reading comprehension.

This approach can have a positive impact on oral language outcomes too, such as a pupil's ability to retell a story or understand spoken vocabulary. The evidence looks promising, but measuring oral language ability reliably can be difficult. It has different dimensions, including both speaking and listening.

Literacy is important across the curriculum, and oral language approaches can support literacy. A small number of studies test whether oral language interventions improve science and maths outcomes. The impact on achievement in these subject-specific tests is not clear.

There may be wider benefits to oral language approaches outside of educational attainment. For example, some studies report improved pupil attitudes, attention, self-regulation or confidence.

Studies that focus on speaking, listening, or a combination of the two, all show positive impacts on attainment. The majority of studies focus on both together. Studies that focus on listening alone

may be particularly effective, but this evidence base is smaller. Wider evidence suggests that some pupils may struggle specifically with spoken language. Schools should consider how they will identify pupils that need additional support around oral language and articulation. It may be helpful to focus on speaking and listening activities separately where needed to meet particular needs.

Behind the average

Closing the disadvantage gap

There is evidence to suggest that pupils from lower socioeconomic backgrounds are more likely to be behind their more advantaged counterparts in developing early language and speech skills, which may affect their school experience and learning later in their school lives.

Oral language interventions can be an effective approach for disadvantaged pupils. In the research evidence, the higher the proportion of disadvantaged pupils taking part, the more effective the intervention seemed to be.

These approaches can be delivered to all pupils, or they can be targeted. Targeted and supplementary interventions can be effective and so could be used to support any disadvantaged pupils who have fallen behind. However, it is important to consider any potential negative impact of delivering teaching outside the classroom, such as missing out on other lessons.

How could you implement in your setting?

Oral language interventions are implemented in different ways throughout the research evidence. It seems that different approaches can be effective.

Integration with curriculum: Most interventions in this evidence base were integrated within the usual curriculum. When integrated, this was almost always in literacy teaching, and very often in the context of story reading (especially with younger children). There were some examples of interventions embedded in other subjects, including science. Some interventions were delivered outside of the usual classroom setting, in or out of school hours.

It is important to ensure that oral language activities are linked to the wider curriculum. For interventions that target vocabulary, this works best when they are related to content currently being studied in school and involve active and meaningful use of any new vocabulary.

Interactions: Interventions often involve spoken interaction between the practitioner and pupils or pupils and their peers. In group or paired work, pupils sometimes share their thought processes, develop ideas, or consolidate vocabulary. The practitioner has a key role in setting up productive peer discussions. In some interventions, oral language was used by a pupil independently, for example to support metacognition.

Groupings: Lots of the studies test oral language interventions with large groups (more than six pupils, up to a whole class). Some studies take a targeted approach, delivering an intervention to a particular pupil group based on need, for example low-attaining or disadvantaged pupils.

There are EEF resources that schools might find useful in supporting [effective oral language practice](#) and [building strong readers](#).

Whilst this strand includes evidence from children in the Early Years, additional EEF resources offer further support for that age group:

When introducing new approaches, schools should consider implementation. For more information see [A School's Guide to Implementation](#).

What does it cost

Overall, the median costs of implementing oral language interventions are estimated as very low. The costs associated with oral language interventions largely arise from books, resources, and training, the majority of which are start-up costs.

Whilst the median cost estimate for oral language interventions is very low, the option to provide training for staff means that costs can range from very low to moderate.

How secure is the evidence?

The security of the evidence around oral language interventions is rated as high. 188 studies were identified. Overall, the topic lost one padlock because a large percentage of the studies were not independently evaluated. Evaluations conducted by organisations connected with the approach – for example, commercial providers, typically have larger impacts, which may influence the overall impact of the strand.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Parental engagement refers to teachers and schools involving parents in supporting their children's academic learning. It includes:

- approaches and programmes which aim to develop parental skills such as literacy or IT skills;
- general approaches which encourage parents to support their children with, for example reading or homework;
- the involvement of parents in their children's learning activities; and
- more intensive programmes for families in crisis.

Key findings

1. Parental engagement has a positive impact on average of 4 months' additional progress. It is crucial to consider how to engage with all parents to avoid widening attainment gaps.

2. Consider how to tailor school communications to encourage positive dialogue about learning. There is some evidence that personalised messages linked to learning can promote positive interactions.

3. There is a strong evidence base supporting this approach with younger children, but far less research on this approach in secondary schools. It is important to consider how you will maintain parental engagement as children get older.

4. Consider what support you can give to parents to ensure home learning is of high quality. For example, providing practical strategies with tips, support, and resources to assist learning at home may be more beneficial to pupil outcomes than simply gifting a book to pupils or asking parents to provide generic help to their children.

How effective is the approach?

The average impact of the Parental engagement approaches is about an additional four months' progress over the course of a year.

The evidence about *how* to improve attainment by increasing parental engagement is mixed and much less conclusive. There are

examples where combining parental engagement strategies with other interventions, such as extended early years provision, has not been associated with any additional educational benefit. This suggests that developing effective parental engagement to improve their children's attainment is challenging and needs careful monitoring and evaluation.

There is some evidence that supporting parents with their first child will have benefits for siblings.

Parents' aspirations also appear to be important for pupil outcomes, although there is limited evidence to show that intervening to change parents' aspirations will raise their children's aspirations and achievement over the longer term.

The EEF has tested a number of interventions designed to improve pupils' outcomes by engaging parents in different types of skills development. The consistent message from these has been that it is difficult to engage parents in programmes. By contrast, a trial which aimed to prompt greater parental engagement through text message alerts delivered a small positive impact, and at very low cost. Sometimes, parental engagement is an aspect of a multicomponent intervention, as in the case of the promising programme [Success for All](#). However, we do not know how much parental engagement contributes to the effectiveness of this programme.

Behind the average

Closing the disadvantage gap

Disadvantaged pupils are less likely to have access to a space to conduct home learning, or to formal and informal learning activities outside of school. This may make it more difficult for some disadvantaged pupils to learn outside of school. This might explain why attainment levels regress during the summer holidays for some disadvantaged pupils. By designing and delivering effective approaches to support parental engagement, schools and teachers may be able to mitigate this risk, supporting parents to assist their children's learning or their self-regulation, as well as specific skills, such as reading.

However, parental engagement strategies have the risk of increasing attainment gaps if the families that access parental engagement opportunities are primarily from affluent backgrounds. There is some research evidence to suggest that some interventions can widen attainment gaps, but more evidence is needed. It is crucial to consider how parental engagement strategies will support all parents.

While encouraging parents to become directly involved in homework might appear attractive, schools should consider whether parents have the knowledge and skills to provide the right support, particularly at secondary level. Interventions designed to engage parents in homework have generally not been linked to increased attainment. Students who are struggling academically may be more likely to request parental assistance with homework, but parents may be unfamiliar with the most effective teaching methods. As a consequence, it may be more effective to encourage parents to redirect a struggling pupil to their teachers rather than to take on an instructional role.

How could you implement in your setting?

The key mechanism for parental engagement strategies is improving the quality and quantity of learning that takes place in the home learning environment. This is very challenging to implement in practice. Some key elements schools might choose to implement include:

Implementing parental engagement strategies needs to consider potential barriers to parents engaging. For example, is there provision for working parents to engage in short sessions with flexible times – or even through remote engagement where available?

Parental engagement approaches are typically delivered over an extended period (around 18 weeks), as building effective relationships between school and parents requires a sustained effort.

When introducing new approaches, schools should consider implementation. For more information see [A School's Guide to Implementation](#).

What does it cost

Overall, the median costs of implementing parental engagement are estimated as very low. Most costs arise from staff training and development, all of which are more likely to be start-up costs.

Whilst the median cost estimate for parental engagement is very low, the option to include additional ongoing staff training, materials and resources, and additional staff time means that costs can range from very low to moderate.

These cost estimates assume that schools are already paying for technology for communication with parents, and facilities to host any in person meetings. These are all pre-requisite costs of implementing parental engagement strategies, without which the cost is likely to be higher.

How secure is the evidence?

The security of the evidence around parental engagement is rated as high. 124 studies were identified. The topic lost a padlock because a large percentage of the studies were not independently evaluated. Evaluations conducted by organisations connected with the approach – for example, commercial providers, typically have larger impacts, which may influence the overall impact of the strand.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Peer tutoring includes a range of approaches in which learners work in pairs or small groups to provide each other with explicit teaching support, such as:

- cross-age tutoring, in which an older learner takes the tutoring role and is paired with a younger tutee or tutees;
- same-age peer tutoring, in which learners of the same age take on fixed roles as tutor or tutee; and
- reciprocal peer tutoring, in which pupils (usually of the same age) alternate between the role of tutor and tutee.

The common characteristic is that learners take on responsibility for aspects of teaching and for evaluating their success.

Peer assessment involves the peer tutor providing feedback to the tutee relating to their performance and can take different forms, such as reinforcing learning or correcting misunderstandings.

Peer tutoring has some cross-overs with [Collaborative learning](#)

approaches, which is its own topic in the Toolkit. Other types of tutoring, where an adult is the tutor, include [Small group tuition](#) and [One to one tuition](#).

Key findings

1. Peer tutoring, on average, has a positive impact on both tutors and tutees and may be a cost-effective approach to delivering one to one or small group tuition in a school.

2. Peer tutoring seems most effective when used to review or consolidate learning, rather than introducing new material.

3. Training for staff and tutors is essential for success. It is crucial to allocate sufficient time to train both staff and tutors, to ensure training provides structure to the tutoring, and to identify and implement improvements as the programme progresses.

4. Peer tutoring may be particularly effective for low-attaining pupils in the role of tutee. This approach shows promise for supporting these pupils to catch up with their peers.

How effective is the approach?

Peer tutoring approaches have been shown to have a positive impact on learning, with an average positive effect equivalent to approximately six additional months' progress within one academic year. Studies have identified benefits for both tutors and tutees, and for a wide range of age groups. Though all types of pupils appear to benefit from peer tutoring, there is some evidence that pupils who are low-attaining and those with special educational needs make the biggest gains.

Peer tutoring appears to be particularly effective when pupils are provided with support to ensure that the quality of peer interaction is high: for example, questioning frames to use in tutoring sessions, and training and feedback for tutors. In cross-age peer tutoring some studies have found that a gap of less than three years is optimal, although ensuring that the gap is wide enough so that the work is challenging to the tutee whilst easy enough for the tutor to support them is key. Regular tutoring sessions (4–5 times a week) of up to 10 weeks appear to be more effective than less intensive or longer programmes.

Successful approaches may also have other benefits, such as supporting the social and personal development of pupils and boosting their self-confidence and motivation for learning.

Behind the average

Closing the disadvantage gap

While there is limited evidence that specifically examines pupils from a disadvantaged background, studies have shown that pupils who are low attaining typically receive additional benefits from peer tutoring. Peer-led tutoring

approaches may help pupils to close gaps in their learning by offering targeted, peer-led support to consolidate within class learning, practice skills, and identify and overcome misconceptions. There is also some evidence to suggest that peer-led tutoring can offer tutors the chance to revisit and revise skills, prior knowledge, and develop metacognitive understanding of topics.

Applications and approaches

Impact on tutors

+6

Reciprocal tutoring

+7

How could you implement in your setting?

Peer tutoring relies on close interaction between two or more students with learners taking responsibility for aspects of teaching and for evaluating their success. When implementing peer tutoring approaches, schools should consider how to ensure high quality interactions between pupils. This might include:

- Carefully structured tasks so sessions focus on existing knowledge;
- Training for peer tutors on teaching approaches, such as modelling knowledge, overcoming common misconceptions, feedback and evaluating progress;
- Careful consideration of appropriate pairing of tutors and tutees;
- Providing teaching aids and learning frames to guide tutors on how to structure learning, or the types of questions to ask tutees.

Peer tutoring interventions are typically delivered over four to ten week intensive blocks. Approaches may involve cross-age or same-age tutoring usually in pairs. Approaches may be based on a fixed tutee and tutor relationship, while others may be reciprocal.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School’s Guide to Implementation](#).

What does it cost

The average cost of peer tutoring is expected to be very low. The cost to schools is largely based on teacher training and learning resources. Implementing peer tutoring will also require a moderate amount of staff time, compared with other approaches.

Alongside time and cost, school leaders should consider how to maximise the quality of peer tutoring interactions and ensure sufficient time is allocated to identify and implement improvements to approaches. When utilising programmes, school leaders should assess the quality and strength of evidence behind them.

How secure is the evidence?

The security of the evidence around peer tutoring is rated as high. 145 studies were identified that meet the inclusion criteria of the Toolkit. The topic lost a padlock because a large percentage of the studies were not independently evaluated. Evaluations conducted by organisations connected with the approach – for example, commercial providers, typically have larger impacts, which may influence the overall impact of the strand.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Performance pay schemes aim to create a direct link between teacher pay and the performance of their class in order to incentivise better teaching and so improve pupil outcomes.

A distinction can be drawn between awards, where improved performance leads to a higher permanent salary, and payment by results, where teachers get a bonus for higher test scores in a particular school year.

Approaches also differ in how performance is measured and how closely those measures are linked to outcomes for learners. In some schemes, students' test outcomes are the sole factor used to determine performance pay awards. In others, performance judgements can also include information from lesson observations

or feedback from pupils, or be left to the discretion of the headteacher. Some performance schemes incentivise individuals, others incentivise groups of teachers.

Key findings

1. The impact of performance pay is low (+1 month), schools might consider other, more cost effective, ways to improve teacher performance, such as high quality continuing professional development.

2. Given the lack of evidence that performance pay significantly improves the quality of teaching, resources may be better targeted at developing existing teachers.

3. Implementing performance related pay can narrow the focus of teachers to particular groups or particular measures, so care should be taken to try and minimise undesirable effects.

How effective is the approach?

The results of rigorous evaluations, such as those with experimental trials or with well-controlled groups, suggest that the average impact of performance pay schemes has been low (+1 months' progress). The use of loss aversion – where awards have to be paid back if student results fall below a certain level – are less widespread in many education systems, but where these approaches have been trialled evaluations have found some larger impacts.

There is some evidence that impact may also be greater in developing countries. Overall, evaluations of a number of performance pay schemes in the USA, where the approach is also known as 'merit pay', have been unable to find a clear link with improved pupil learning outcomes.

There are some concerns that performance pay schemes can create unintended consequences. For example, that they may encourage teachers to focus only on tested outcomes, leading to a narrowing of the curriculum, or to focus on groups of pupils near a particular assessment threshold.

A related approach that has shown promise is using bonuses or enhanced pay to attract teachers to challenging schools.

Overall, approaches which simply assume that incentives will make teachers work more effectively are not well supported by existing evidence.

Behind the average

Closing the disadvantage gap

Studies in England have shown that pupils in disadvantaged communities are less likely have access to high performing schools and are more likely to attend schools staffed by teachers without qualified teacher status or with fewer years of experience.

Limited access to high-quality teaching is likely to be a key contributor to the disadvantage attainment gap, with research consistently demonstrating the positive impact of high-quality teaching on pupil attainment. Pay may be utilised as part of a strategy to attract and retain experienced, qualified and specialist teachers to schools with higher numbers of pupils eligible for the pupil premium.

The small impact of performance pay, specifically, may mean that schools and policymakers consider other approaches to attracting high quality teachers to schools in disadvantaged communities.

How could you implement in your setting?

There are several different approaches to performance pay including bonuses, loss aversion and enhanced pay.

Key elements of performance pay approaches might include:

It is crucial that strategies that reward teacher performance also include support for teachers to develop and careful strategies to mitigate the risk of focusing only on the measures used to distribute performance related pay.

What does it cost

Overall, the median costs of implementing performance pay are estimated as low. The costs associated with performance pay arise from either paying bonuses to staff, or increases in staff salaries, all of which are recurring

costs.

Whilst the median cost estimate for performance pay is low, the range in the amount some schools pay to staff as bonuses means that the costs can range from very low to moderate. For example, bonuses can range from £300 to £10,000 per teacher (or between £12 and £400 per pupil across a class of 25).

In England, performance pay generally means schools using staff performance to allocate teacher progression within existing pay ranges. Increases according to this scale range from around £1000 to £3000 (or between £40 to £120 per pupil across a class of 25).

These cost estimates assume that schools apply performance pay schemes to the bonuses or salaries of classroom teachers. Applying performance related pay to the roles of senior leaders and headteachers is likely to lead to higher overall costs.

Alongside cost, school leaders should consider how to retain high-quality teaching staff through a strong culture of professional support and development. School leaders should carefully monitor the impact of performance pay approaches and avoid schemes that can lead to unintended consequences, such as a greater attention on specific pupil groups closer to assessment thresholds at the detriment to other students.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School's Guide to Implementation](#).

How secure is the evidence?

The security of the evidence around performance is rated as very low. 27 studies were identified that meet the inclusion criteria for the Toolkit. The topic lost padlocks because:

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting

What is it?

Phonics is an approach to teaching some aspects of literacy, by developing pupils’ knowledge and understanding of the relationship between written symbols and sounds. This involves the skills of hearing, identifying and using the patterns of sounds or phonemes to read written language. The aim is to systematically teach pupils the relationship between these sounds and the written spelling patterns, or graphemes, which represent them. Phonics emphasises the skills of decoding new words by sounding them out and combining or ‘blending’ the sound-spelling patterns.

Key findings

1. Phonics has a positive impact overall (+5 months) with very extensive evidence and is an important component in the development of early reading skills, particularly for children from disadvantaged backgrounds.
2. The teaching of phonics should be explicit and systematic to support children in making connections between the sound patterns they hear in words and the way that these words are written.
3. The teaching of phonics should be matched to children’s current level of skill in terms of their phonemic awareness and their knowledge of letter sounds and patterns (graphemes).
4. Phonics improves the accuracy of the child’s reading but not necessarily their comprehension. It is important that children are successful in making progress in all aspects of reading including comprehension, the development of vocabulary and spelling, which should also be taught explicitly.

How effective is the approach?

The average impact of the adoption of phonics approaches is about an additional five months’ progress over the course of a year.

Phonics approaches have been consistently found to be effective in supporting younger pupils to master the basics of reading, with an average impact of an additional five months' progress. Research suggests that phonics is particularly beneficial for younger learners (4–7 year olds) as they begin to read. Teaching phonics is more effective on average than other approaches to early reading (such as whole language or alphabetic approaches), though it should be emphasised that effective phonics techniques are usually embedded in a rich literacy environment for early readers and are only one part of a successful literacy strategy.

While there have been fewer studies examining phonics with older readers, there is evidence that it can be a positive approach. With any reading intervention, careful diagnosis is required on the difficulties that the reader is experiencing, regardless of age. If an older reader is struggling with decoding, phonics approaches will still be appropriate. Where readers are struggling with vocabulary or comprehension, other interventions may be more appropriate.

There is some variation in impact between different phonological approaches. Synthetic phonics approaches have higher impacts, on average, than analytic approaches. Analytic phonics approaches has also been studied less overall (only 9 studies). The small number of analogic phonics approaches identified in this review (6 studies) have a negative impact on average.

Behind the average

Closing the disadvantage gap

Studies in England have shown that pupils eligible for free school meals typically receive similar or slightly greater benefit from phonics interventions and approaches. This is likely to be due to the explicit nature of the instruction and the intensive support provided.

It is possible that some disadvantaged pupils may not develop phonological awareness at the same rate as other pupils, having been exposed to fewer words spoken and books read in the home. Targeted phonics interventions may therefore improve decoding skills more quickly for pupils who have experienced these barriers to learning.

How could you implement in your setting?

Phonics approaches aim to quickly develop pupils' word recognition and spelling through developing pupils' ability to hear, identify and manipulate phonemes (the smallest unit of spoken language), and to teach them the relationship between phonemes and the graphemes (written letters or combinations of letters) that represent them. Successfully implementing a phonics might involve:

Good implementation of phonics programmes will also consider pupils wider reading skills and will identify where pupils are struggling with aspects of reading other than decoding that might be targeted through other approaches such as the explicit teaching of reading comprehension strategies.

Where phonics is delivered as an intervention targeted at specific pupils, regular sessions (up to four times a week), of 30 minutes or so over a period of up to 12 weeks appear to be the most successful structure.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School’s Guide to Implementation](#).

What does it cost

Overall, the median costs of implementing a phonics intervention are estimated as very low. The costs associated with teaching phonics arise from the need for specific resources and professional training, the majority of which are initial start-up costs paid during the first year of delivery.

Whilst the median cost estimate for phonics programmes is very low, the range of prices between available programmes and the option to purchase additional ongoing training and support for teaching staff means that costs can range from very low to low. Evidence suggests that the effectiveness of phonics is related to the pupil’s stage of reading development, so it is important that teachers have professional development in effective assessment as well as in the use of particular phonics techniques and materials.

These cost estimates assume that schools are already paying for staff salaries to deliver interventions, facilities to host lessons, and basic stationary materials for staff and pupils. These are all pre-requisite costs of implementing a phonics intervention, without which the cost is likely to be higher.

How secure is the evidence?

The security of the evidence around phonics is rated as very high. 121 studies were identified that meet the inclusion criteria of the Toolkit.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Physical activity refers to approaches that engage pupils in sports, dance, or any kind of physical exercise.

This might be through organised after school activities or a programme organised by a local sporting club or association. Sometimes sporting activity is used as a means to encourage young people to engage in additional

learning activities, such as football training at a local football club combined with study skills, ICT, literacy, or mathematics lessons.

Physical activity has important benefits in terms of health, wellbeing and physical development. These benefits have important value in themselves, however, this Toolkit entry focuses on the benefits of physical activity for core academic attainment particularly literacy and mathematics.

Key findings

There is a small positive impact of physical activity on academic attainment (+2 month). While this evidence summary focuses on the link between physical activity and academic performance, it is crucial to ensure that pupils access to high quality physical activity for the other benefits and opportunities it provides.

Impact on attainment varies considerably between different interventions, and participation in sports does not straightforwardly transfer to academic learning. It is likely that the quality of the programme and the emphasis on or connection with academic learning may make more difference than the specific type of approach or activities involved.

Planned extra-curricular activities which include short, regular, and structured teaching in literacy and mathematics (either tutoring or group teaching) as part of a sports programme, such as an after-school club or summer school, are more likely to offer academic benefits than sporting activities alone.

There is some evidence that involvement in extra-curricular sporting activities may increase pupil attendance and retention.

How effective is the approach?

The average impact of the engaging in physical activity interventions and approaches is about an additional two month's progress over the course of a year.

The variability in effects suggests that the quality of the programme and the emphasis on, or connection with, academic learning may make more difference than the specific type of approach or sporting

activities involved. Participating in sports and physical activity is likely to have wider health and social benefits.

There are wider benefits from regular physical activity in terms of physical development, health and wellbeing as well as other potential benefits have been reported such as improved attendance.

Behind the average

Closing the disadvantage gap

Pupils from disadvantaged backgrounds may be less likely to be able to benefit from sport clubs and other physical activities outside of school due to the associated financial costs (e.g. equipment). By providing physical activities free of charge, schools give pupils access to benefits and opportunities that might not otherwise be available to them.

When considering hosting sports activities that may require contributions from parents, schools should consider whether places could be provided free of charge or subsidised for disadvantaged pupils.

How could you implement in your setting?

The way in which physical activity influences academic attainment is unclear. It may be the actual physical exercise that influences attainment, or supplementary learning activities that have been integrated into physical activity in the studies located in the Toolkit. Some ways that schools implement physical activity include:

More information on how Physical Activity can promote physical development and outcomes for pupils in the Early Years is found in the [Early Years Evidence Store](#).

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School’s Guide to Implementation](#).

What does it cost

Overall, the median costs of implementing physical activity are estimated as very low. The costs associated with physical activities arise from training for staff leading programmes, and any additional resource or equipment required, the majority of which are start-up costs.

Whilst the median cost estimate for physical activity is very low, the possible addition of extra staff salaries and facilities means that costs can range from very low to moderate.

These cost estimates assume that schools are already paying for school sports facilities and some basic materials and equipment. These are all pre-requisite costs of implementing physical activities, without which the cost is likely to be higher.

How secure is the evidence?

The security of the evidence around physical activity is rated as high. 136 studies were identified that meet the inclusion criteria for the Toolkit. The topic lost a padlock because a large percentage of the studies were not independently evaluated. Evaluations conducted by organisations connected with the approach – for example, commercial providers, typically have larger impacts, which may influence the overall impact of the strand.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Reading comprehension strategies focus on the learners’ understanding of written text. Pupils learn a range of techniques which enable them to comprehend the meaning of what they read. These can include: inferring meaning from context; summarising or identifying key points; using graphic or semantic organisers; developing questioning strategies; and monitoring their own comprehension and then identifying and resolving difficulties for themselves (see also [metacognition and self-regulation](#)).

Strategies are often taught to a class and then practiced in pairs or small groups (see also [collaborative learning approaches](#)).

Key findings

1. Reading comprehension strategies are high impact on average (+6 months). Alongside phonics it is a crucial component of early reading instruction.
2. It is important to identify the appropriate level of text difficulty, to provide appropriate context to practice the skills, desire to engage with the text and enough challenge to improve reading comprehension.
3. Effective diagnosis of reading difficulties is important in identifying possible solutions, particularly for older struggling readers. Pupils can struggle with decoding words, understanding the structure of the language used, or understanding particular vocabulary, which may be subject-specific.
4. A wide range of strategies and approaches can be successful, but for many pupils they need to be taught explicitly and consistently.
5. It is crucial to support pupils to apply the comprehension strategies independently to other reading tasks, contexts and subjects.

How effective is the approach?

The average impact of reading comprehension strategies is an additional six months' progress over the course of a year. Successful reading comprehension approaches allow activities to be carefully tailored to pupils' reading capabilities, and involve activities and texts that provide an effective, but not overwhelming, challenge.

Many of the approaches can be usefully combined with Collaborative learning techniques and Phonics activities to develop reading skills. The use of techniques such as graphic organisers and drawing pupils' attention to text features are likely to be particularly useful when reading expository or information texts.

There are some indications that approaches involving digital technology can be successful in improving reading comprehension (although there are relatively few studies in this area), particularly when they focus on the application and practice of specific strategies and the use of self-questioning skills.

Supporting struggling readers is likely to require a coordinated effort across the curriculum and a combination of approaches that include phonics, reading comprehension and oral language approaches. No particular strategy should be seen as a panacea, and careful diagnosis of the reasons why an individual pupil is struggling should guide the choice of intervention strategies.

Behind the average

Closing the disadvantage gap

Studies in England have shown that pupils eligible for free school meals may receive additional benefits from being taught how to use reading comprehension strategies. However, the UK evidence base is less extensive than the global average, and UK studies show lower impact for all pupils.

Reading comprehensions strategies involve the teaching of explicit approaches and techniques a pupil can use to improve their comprehension of written text. Many learners will develop these approaches without teacher guidance, adopting the strategies through trial and error as they look to better understand texts that challenge them. However, we know that on average, disadvantaged children are less likely to own a book of their own and read at home with family members, and for these reasons may not acquire the necessary skills for reading and understanding challenging texts.

How could you implement in your setting?

Reading comprehension strategies work through a number of different mechanisms – all focused on improving the understanding of meaning of text effectively. Common elements include:

Reading comprehension strategy interventions are typically delivered between one to three terms of a school year, either by teachers within class settings, or by teaching assistants with smaller groups.

Evidence suggests that reading comprehension approaches need to be tailored to pupils' current reading capabilities, so it is important that teachers receive professional development in effective diagnosis as well as training in the use of particular techniques and materials.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School's Guide to Implementation](#).

What does it cost

The average cost of reading comprehension strategies is estimated as very low. The cost to schools is largely based on training and professional development, books and learning resources, the majority of which are initial start-up costs paid during the first year of delivery. Whilst the median cost estimate for reading comprehension programmes is very low, the range of prices between available programmes and the option to purchase additional ongoing training and support for teaching staff means that costs can range from very low to low.

Effective teaching of reading comprehension strategies will also require a moderate amount of staff time, compared with other approaches. Alongside time and cost, school leaders should consider how to develop teachers' ability to use specific techniques for particular pupils' needs and ensure they use texts that provide an effective challenge to readers.

How secure is the evidence?

The security of the evidence around reading comprehension strategies is rated as high. 141 studies were identified that met the inclusion criteria for the Toolkit. Overall, the topic lost one padlock because a large percentage of the studies were not independently evaluated. Evaluations conducted by organisations connected with the approach – for example, commercial providers, typically have larger impacts, which may influence the overall impact of the strand.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Reducing class size is an approach to managing the ratio between pupils and teachers, as it is suggested that the range of approaches a teacher can employ and the amount of attention each student will receive will increase as the number of pupils per teacher becomes smaller.

Key findings

1. Reducing class size has a small positive impacts of +2 month, on average. The majority of studies examine reductions of 10 pupils. Small reductions in class size (for example, from 30 to 25 pupils) are unlikely to be cost-effective relative to other strategies.
2. There is some evidence for additional benefits of smaller class sizes with younger children, so smaller class sizes may be a more effective approach during the early stages of primary school.
3. Smaller classes only impact upon learning if the reduced numbers allow teachers to teach differently – for example, having higher quality interactions with pupils or minimising disruption.

4. The gains from smaller class sizes are likely to come from the increased flexibility for organising learners and the quality and quantity of feedback the pupils receive (see [Feedback](#)).

5. As an alternative to reducing class sizes, it may be possible to change the deployment of staff (both teachers and teaching assistants) so that teachers can work more intensively with smaller groups (see [Small group tuition](#)).

How effective is the approach?

The average impact for reducing class size is around 2 months additional progress over the course of an academic year. The evidence in this area is very limited, so should be treated with caution.

The key issue appears to be whether the reduction is large enough to permit the teacher to change their teaching approach when working with a smaller class and whether, as a result, the pupils change their learning behaviours. If no change occurs then, perhaps unsurprisingly, learning is unlikely to improve. When a change in teaching approach does accompany a class size reduction (which appears hard to achieve until classes are smaller than about 20) then benefits on attainment can be identified, in addition to improvements on behaviour and attitudes.

Behind the average

Closing the disadvantage gap

International research evidence suggests that reducing class size can have positive impacts on pupil outcomes when implemented with socioeconomically disadvantaged pupil populations. Some studies also have also found that smaller class sizes in primary schools can have a greater positive impact on disadvantaged pupils than their peers.

In the UK, there is some indicative evidence to suggest that reception and Key Stage 1 pupils with lower prior attainment and from lower socioeconomic backgrounds may benefit from small classes, although the class size threshold at which this impact is identifiable varies between literacy and maths, and potentially also geographical area.

How could you implement in your setting?

The evidence suggests that significant effects of reducing class size are not seen until the number of pupils has decreased substantial (to fewer than 20 or even 15 pupils). Crucially, a reduction in class size is only likely to be effective if it permits teachers to change their teaching approach to the extent that this changes the learning behaviours of pupils. High quality implementation of reducing class size might consider:

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School's Guide to Implementation.](#)

What does it cost

The costs associated with reducing class sizes are very high, as additional staff would be required to reduce class sizes.

In 2020/21, the average salary for a primary school teacher in England was £36,900. The average salary for a secondary school teacher was £39,900.

This estimate does not consider the potential cost of sourcing facilities to host the additional lessons created through reducing class size. Space to host lessons is therefore a pre-requisite to reducing class size, without which the costs are likely to be much higher.

How secure is the evidence?

The security of the evidence around reducing class size is rated as very limited. 41 studies were identified. Overall, the topic lost three additional padlocks because:

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Repeating A Year (also known as “grade retention”, “non-promotion”, or “failing a grade”) describes the process by which pupils who do not reach a given standard of learning at the end of a year are required to join a class of younger students the following academic year. For students at secondary school level, repeating a year is usually limited to the particular subject or classes that a student has not passed.

Repeating a year is very rare in the UK, but it is relatively common in the USA, where the No Child Left Behind Act (2002) recommended that students be required to demonstrate a set standard of achievement before progressing to the next grade level. Students can also be required to repeat a year in some European countries including Spain, France, and Germany. In some countries, such as Finland, pupils can repeat a year in exceptional circumstances, but this decision is made collectively by teachers, parents, and the student, rather than on the basis of end of year testing.

Key findings

1. Requiring pupils to repeat a year has a negative impact on average. Negative effects are rare for educational interventions, so the extent to which pupils who repeat a year make less progress is striking.

2. Negative effects are disproportionately greater for disadvantaged pupils, for pupils from ethnic minorities, and for pupils who are relatively young in their year group.
3. Where pupils are not achieving expected outcomes, alternative interventions might provide intensive support that may make repeating a school year unnecessary, e.g [One to one tuition](#).
4. Negative effects tend to increase with time and repeating more than one year significantly increases the risk of students dropping out of school.

How effective is the approach?

The average impact of a pupil repeating a year is about three months' less progress over the course of a year than if the same pupil had not repeated the year, when compared with similar pupils.

In addition, studies consistently show greater negative effects for students from disadvantaged backgrounds, suggesting that the practice is likely to increase educational inequality.

Repeating a year is also likely to lead to greater negative effects when used at secondary school, for students from ethnic minorities, or for pupils who are relatively young in their year group (often referred to as 'summer born' pupils in the US and European literature).

Pupils who repeat a year make an average of three months' less academic progress over the course of a year than pupils who move on. In addition, studies suggest that students who repeat a year are unlikely to catch up with peers of a similar level who move on, even after completing an additional year's schooling. Studies also suggest that students who repeat a year are more likely to drop out of school prior to completion.

Although the overall average impact is negative, some studies suggest that in individual circumstances some students can benefit, particularly in the short term. However, it does not appear to be easy to identify which students will benefit, suggesting that repeating a year is a significant risk.

Behind the average

Closing the disadvantage gap

Research into the socioeconomic “attainment gap” suggests that disadvantaged pupils have, on average, lower attainment than their more advantaged peers. As a result, disadvantaged pupils are more likely to be asked to repeat years or grades, which is likely to entrench underachievement and lead to a lack of confidence and motivation associated with school.

How could you implement in your setting?

Given the typical negative impact of making pupils repeat a year, it is not recommended that schools adopt this approach. Some of the reasons that repeating a year might have a negative impact include pupils feeling stigmatised for failure and being in a class with younger pupils. In exceptional circumstances it may be beneficial for a pupil to repeat a year, such as if they have missed schooling through illness or for other legitimate reasons. If this is the case, it is crucial to:

The average overall negative effects of repeating a year on pupils’ learning suggests that teachers, schools, and parents should consider other approaches in supporting pupils to catch up with peers and reach appropriate levels for their year group or grade level. For example, [One-to-One Tuition](#) could be used as an intervention to target gaps in understanding and provide additional support.

What does it cost

Annual funding for school places, as determined by the National Funding Formula for the 2021 – 22 financial year, is likely to fall between £3,000 and £4,500 per pupil for Primary school children, and between £4,500 and £6,000 for Secondary school pupils. The costs associated with one individual pupil repeating a year are therefore very high.

How secure is the evidence?

The security of the evidence around repeating a year is rated as low. 71 studies were identified that meet the inclusion criteria of the Toolkit. Overall, the topic lost two additional padlocks because:

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

School uniform is the clothing that pupils are required to wear at school. Uniforms differ from the very formal and detailed (such as requiring blazers and specifying other items of clothing) to more informal and brief (such as involving just a school sweatshirt). Schools vary as to how strictly a uniform policy is enforced and whether it includes other aspects of pupils' appearance.

Key findings

1. The current evidence base on school uniform and academic outcomes is extremely weak. The limited evidence base means that an overall impact in months progress is not communicated.
2. Wearing a uniform is not, on its own, likely to improve learning, but can be successfully incorporated into a broader school improvement process which includes the development of a school ethos and the improvement of behaviour and discipline.
3. Staff commitment to upholding and consistently maintaining a uniform policy is crucial to successful implementation.
4. If a uniform policy is in place, it is important to consider how to support families that may not be able to afford uniform.

How effective is the approach?

The lack of studies identified that tested school uniform approaches mean that there is not enough security to communicate a month's progress figure. School uniform policies are often implemented alongside other improvement measures, which make it particularly challenging to measure the impact of uniform interventions alone. As a result, there is not enough security in the evidence to communicate a month's progress figure.

There is a belief in some countries that school uniform supports the development of a whole school ethos and therefore supports discipline and motivation. Some also believe that a uniform promotes social equity. However, there is little robust evidence that introducing a school uniform will, by itself, improve academic performance, behaviour, or attendance.

There is a general belief in the UK that school uniform leads to improvements in pupils' behaviour. It is important to remember that improved behaviour, on its own, does not necessarily lead to better learning, though it may be an important precondition (see [Behaviour interventions](#)).

Closing the disadvantage gap

Pupils from lower socioeconomic households are less likely to be able to afford the cost of school uniforms.

Schools intending to change their school uniform policy should therefore consider what provision can be made to cover the costs of uniform changes for disadvantaged pupils.

How could you implement in your setting?

School uniform policies are thought to complement the development and support of a whole school culture and approach, which in turn may assist pupil discipline and motivation.

Core components of a school uniform policy might include staff holding high expectations of pupils' behaviour and that their attire reflects the values and culture of the school which the pupils reflect.

The implementation of a school uniform policy likely to be a significant cost for parents. It would not be advisable to change school uniform requirements regularly, as this may further disadvantage children and families of lower socioeconomic households.

What does it cost

The costs associated with introducing a school uniform are very low for a school, and mainly depend on parents buying the specified clothing instead of other clothes the child would wear. Some schools may subsidise or pay for the school uniform of children from low-income families, which is likely to slightly increase the cost associated

with introducing a school uniform.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School’s Guide to Implementation](#).

How secure is the evidence?

There is extremely limited evidence on the impact of school uniform on attainment outcomes. For topics with extremely low evidence, a month’s progress figure is not displayed. Only 7 studies were found that met the inclusion criteria for the Toolkit.

What is it?

‘Setting’ or ‘streaming’ refer to a variety of approaches by which pupils with similar levels of current attainment are consistently grouped together for lessons.

- ‘Setting’ usually involves grouping pupils in a given year group into classes for specific subjects, such as mathematics and English, but not across the whole curriculum.
- ‘Streaming’ (also known as ‘tracking’ in some countries) usually involves grouping pupils into classes for all or most of their lessons, so that a pupil is in the same group regardless of the subject being taught.

Pupils in different sets or streams sometimes follow a different curriculum, particularly when different national tests, different examination levels or different types of academic and vocational qualifications are available. The aim of setting and streaming approaches is to enable more effective and efficient teaching by narrowing the range of pupil attainment in a class.

Although these practices are sometimes described as ‘ability grouping’, we refer here to ‘attainment’ rather than ‘ability’, as schools generally use measures of current performance, rather than measures of ability, to group pupils. Setting and streaming are combined in this Toolkit entry because these practices are similar in that they separate pupils into whole classes of similar levels of attainment.

For evidence on the impact of grouping pupils by attainment within classes, see [within class attainment grouping](#). Other types of attainment grouping, such as grouping by attainment across year groups, and teaching high attaining pupils with older year groups, are not covered in the Toolkit as they are less commonly used.

In the UK, setting and streaming are more common in secondary school than in primary school.

Key findings

1. The impact of setting and streaming is 0 months progress, on average, with worse outcomes for low attaining pupils. The evidence around setting and streaming is limited. Schools may consider other approaches to targeting learning effectively for pupils (e.g. small group or one to one tuition).
2. If schools choose to use setting and streaming it is vital to consider how the approach will enable more effective teaching for all pupils, including lower attaining pupils. For example, carefully considering how to allocate teachers appropriately to different sets.
3. It is important to ensure that all pupils follow a challenging curriculum, including lower attaining pupils. Ensuring flexibility in grouping arrangements, and regular monitoring of learning will minimise the risk of misallocation for pupils that learn at different rates.
4. Making informed choices about the allocation of pupils to groups is important. For example, there is some evidence of pupils being disadvantaged by their relative age within a year group or through subconscious bias on the basis of race or class.

How effective is the approach?

On average, pupils experiencing setting or streaming make similar progress to pupils taught in mixed attainment classes. The evidence suggests that setting and streaming has a small negative impact on low attaining learners, and a small positive impact for higher attaining pupils. There are exceptions to this pattern, with some variation between studies. Overall, the effects are negligible, and it appears that setting or streaming is not an effective way to raise attainment for most pupils.

Setting or streaming may also have an impact on wider outcomes such as confidence. Some studies from the broader evidence base conclude that grouping pupils on the basis of attainment may have longer term negative effects on the attitudes and engagement of

low attaining pupils, for example, by discouraging the belief that their attainment can be improved through effort.

One of the challenges of attainment grouping is ensuring that pupils are correctly allocated to groups. Some studies from the UK suggest that misallocation is a particular problem for pupils from disadvantaged backgrounds, who are at greater risk of misallocation to lower attaining groups, and the negative impact which can accompany this.

Behind the average

Closing the disadvantage gap

Evidence suggests that disadvantaged pupils may suffer from lower teacher expectations which increases their chances of being placed in lower sets or streams. Pupils in lower sets and streams are more likely to be taught by less experienced and qualified teachers, and often develop a lack of confidence in their own capabilities.

Setting or streaming may construct negative self-fulfilling prophecies for disadvantaged pupils, whereby their chances of improving attainment and experiencing success is hampered by the combination of lower teacher expectations and between class stratification.

There is also some evidence that pupils from disadvantaged backgrounds are more likely to be misallocated to lower sets.

Applications and approaches

Impact for pupils with low prior attainment

-3

Impact for pupils with high prior attainment

+1

How could you implement in your setting?

Setting and streaming aims to improve pupil outcomes by ensuring that class content is well-targeted to pupil needs, and that pupils in lower attaining groups get appropriate support. Given the weakness of the evidence base, strong implementation will consider how to mitigate the potential risks of setting and streaming for low attaining pupils, it might include:

- Careful consideration of teacher allocation between sets to ensure that low attaining pupils receive high quality teaching.
- The use of continued monitoring to ensure that pupils are in appropriate sets and are not misallocated.
- Flexible practices that allow pupils to move between sets.

Approaches to setting and streaming are typically delivered over the course of an academic year, with grouping often being informed by pupils' attainment outcomes from the previous school year. Some schools may however move pupils between sets or streams during the school year.

Moving either from mixed ability grouping to setting and streaming or in the other direction is a substantial change that will need a careful implementation process. When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School's Guide to Implementation.](#)

What does it cost

Grouping by attainment is an organisational strategy which has few, if any, financial costs associated with it.

Additional resources may be needed to support different groups. Overall the costs are estimated as very low.

How secure is the evidence?

The security of the evidence around setting and streaming is rated as very limited. 58 studies were identified that meet the inclusion criteria for the Toolkit. The topic lost padlocks because:

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Small group tuition is defined as one teacher, trained teaching assistant or tutor working with two to five pupils together in a group. This arrangement enables the teaching to focus exclusively on a small number of learners, usually in a separate classroom or working area. Intensive tuition in small groups is often provided to support lower attaining learners or those who are falling behind, but it can also be used as a more general strategy to ensure effective progress, or to teach challenging topics or skills.

Key findings

1. Small group tuition has an average impact of four months' additional progress over the course of a year.
2. Small group tuition is most likely to be effective if it is targeted at pupils' specific needs. Diagnostic assessment can be used to assess the best way to target support.
3. One to one tuition and small group tuition are both effective interventions. However, the cost effectiveness of teaching in small groups indicates that greater use of this approach may be worthwhile.
4. Providing training to the staff that deliver small group support is likely to increase impact.
5. Additional small group support can be effectively targeted at pupils from disadvantaged backgrounds, and should be considered as part of a school's pupil premium strategy.

How effective is the approach?

The average impact of the small group tuition is four additional months' progress, on average, over the course of a year.

Evidence shows that small group tuition is effective and, as a rule of thumb, the smaller the group the better. Some studies suggest that greater feedback from the teacher, more sustained the engagement in smaller groups, or work which is more closely matched to learners' needs explains this impact. Once group size increases above six or seven there is a noticeable reduction in effectiveness.

Although the rule "the smaller the better" there is some variability in impact within the existing evidence. For example, in reading, small group teaching can sometimes be more effective than either one to one or paired tuition. It may be that in these cases reading practice can be efficiently organised so that all the pupils stay fully engaged as each take their turn, such as in Guided Reading.

The variability in findings suggests two things. First, the quality of the teaching in small groups may be as, or more important than, the precise group size (there is evidence of the benefits of staff professional development on pupil outcomes). Second, it is important to evaluate the effectiveness of different arrangements, as the specific subject matter being taught and composition of the groups may influence outcomes.

Given the uncertainty and the lower cost, small group tuition may be a sensible approach to trial before considering to one to one tuition.

Behind the average

Closing the disadvantage gap

Studies in England have shown that pupils eligible for free school meals typically receive additional benefits from small group tuition.

Small group tuition approaches can support pupils to make effective progress by providing intensive, targeted academic support to those identified as having low prior attainment or at risk of falling behind. The approach allows the teacher to focus on the needs of a small number of learners and provide teaching that is closely matched to pupil understanding. Small group tuition offers an opportunity for greater levels of interaction and feedback compared to whole class teaching which can support pupils to overcome barriers to learning and increase their access to the curriculum.

How could you implement in your setting?

Small group tuition has an impact by providing additional support that is targeted at pupil needs. The reduction in the ratio of pupils to teacher compared to a regular classroom setting also allows for closer interaction between educators and pupils. When implementing small group tuition, schools should consider how to ensure that these active ingredients have a positive impact through:

Small group tuition may be delivered by teachers, trained teaching assistants, academic mentors or tutors. Interventions are typically delivered over an extended period, often over the course several weeks or terms, to a small group of between two to five pupils.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School's Guide to Implementation](#).

What does it cost

The average cost of small group tuition is low. The costs to schools are largely based on additional salary costs and learning resources, the majority of which are recurring costs. Through the National Tutoring Programme Year 1 (2020–21), schools could purchase subsidised in person or online tutoring sessions for 1:3 or 1:2 groups in 15hr blocks for an average cost of £70-£100 per pupil, with lower costs for higher pupil-tutor ratios and online delivery.

When delivering teacher or TA-led small group tuition, implementation is likely to require a large amount of staff time compared with whole class approaches. Given the lower costs, small group tuition may be a sensible approach to trial before considering one to one tuition. See [one to one tuition](#).

Alongside time and cost, school leaders should consider using providers with a track record of effectiveness. To increase the impact of small group approaches, school leaders should consider professional development for teachers, TAs, and tutors to support high-quality tutoring practices.

How secure is the evidence?

The security of the evidence around small group tuition is rated as moderate. 62 studies were identified that met the inclusion criteria of the Toolkit. The topic lost an additional padlock because a large percentage of the studies were not independently evaluated:

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Social and emotional learning (SEL) interventions seek to improve pupils' decision-making skills, interaction with others and their self-management of emotions, rather than focusing directly on the academic or [cognitive](#) elements of learning.

SEL interventions might focus on the ways in which students work with (and alongside) their peers, teachers, family or community.

Three broad categories of SEL interventions can be identified:

- School-level approaches to developing a positive school ethos, which also aim to support greater engagement in learning;

- Universal programmes which generally take place in the classroom with the whole class; and
- More specialised programmes which use elements of SEL and are targeted at students with particular social or emotional needs.

Key findings

1. Social and emotional learning approaches have a positive impact, on average, of 3 months' additional progress in academic outcomes over the course of an academic year.

2. The studies in the Toolkit focus primarily on academic outcomes, but it is important to consider the other benefits of SEL interventions. Being able to effectively manage emotions will be beneficial to children and young people even if it does not translate to reading or maths scores.

3. While universal approaches to SEL learning seem to have a slightly greater impacts on average, approaches should not be viewed in opposition, as most schools will want to use a combination of whole class SEL learning, and targeted support for pupils with particular social and emotional needs.

4. The evidence indicates that there is particular promise for approaches that focus on improving social interaction between pupils.

How effective is the approach?

The average impact of successful SEL interventions is an additional three months' progress over the course of a year. Alongside academic outcomes, SEL interventions have an identifiable and valuable impact on attitudes to learning and social relationships in school.

Although SEL interventions are almost always perceived to improve emotional or attitudinal outcomes, not all interventions are equally effective at raising attainment. Improvements appear more likely when SEL approaches are embedded into routine educational practices and supported by professional development and training for staff. In addition, the implementation of the programme and the

degree to which teachers are committed to the approach appear to be important.

Behind the average

Closing the disadvantage gap

Evidence suggests that children from disadvantaged backgrounds have, on average, weaker SEL skills at all ages than their more affluent peers. These skills are likely to influence a range of outcomes for pupils: lower SEL skills are linked with poorer mental health and lower academic attainment.

SEL interventions in education are shown to improve SEL skills and are therefore likely to support disadvantaged pupils to understand and engage in healthy relationships with peers and emotional self-regulation, both of which may subsequently increase academic attainment.

Schools should carefully consider how targeted approaches are deployed to support pupils with additional social or emotional needs. SEL needs will be based on a variety of factors that may not correspond to academic progress and should be carefully monitored.

Applications and approaches

Targeted support

+3

Universal approaches

+4

How could you implement in your setting?

Social and emotional learning is important in and of itself. The mechanism by which approaches have an impact on academic outcomes may include improving engagement in learning or self-regulation skills. If schools are aiming to improve a particular skill, they should carefully consider:

SEL approaches are typically delivered over a pre-specified period if used as a targeted intervention (e.g. length of one term), although they could also be implemented over the course of an academic year (e.g. if purposed with school wide change). When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School’s Guide to Implementation](#).

What does it cost

Overall, the median costs of implementing SEL approaches are estimated as very low. The costs associated with SEL interventions arise from professional training and development for staff, the majority of which are start-up costs.

Whilst the median cost estimate for SEL approaches is very low, the option to purchase additional books, resource and materials, and ongoing training and support means that costs can range from very low to moderate.

How secure is the evidence?

The security of the evidence around SEL approaches is rated as moderate. 119 studies were identified that meet the inclusion criteria for the Toolkit. The topic lost an additional padlock because:

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Summer schools are additional lessons or classes organised during the summer holidays. They are often designed as catch-up programmes, although some do not have an academic focus and concentrate on sports or other non-academic activities. Others have a specific aim, such as supporting pupils at the transition from primary to secondary school or preparing high-attaining pupils for university.

Other approaches to increasing learning time are included in other sections of the Toolkit, such as [homework](#) and [extending school time](#).

Key findings

1. Summer schools have a positive impact on average (three months' additional progress), but are expensive to implement. Providing additional support during the school year may be a more cost effective approach to improving outcomes.
2. Summer school provision that aims to improve learning needs to have an academic component. Summer schools that include an intensive teaching component such as using small group or one to one approaches have higher impacts, on average.
3. Maintaining regular attendance at summer schools can be challenging, in particular for disadvantaged pupils. It is crucial to consider how summer schools will attract and engage pupils to prevent attainment gaps widening.
4. Summer schools that use teachers that are known to the pupils have a higher impact, on average, but may be even more expensive to implement.
5. Summer schools can also provide additional experiences and activities, such as arts or sporting activities. This might be valuable in and of themselves or be used to increase engagement alongside academic support.

How effective is the approach?

On average, evidence suggests that pupils who attend a summer school make approximately three additional months' progress compared to similar pupils who do not attend a summer school.

Greater impact can be achieved when summer schools are intensive, well-resourced, and involve small group or one to one teaching by trained and experienced teachers. It does appear to be an advantage to have teachers who are known to the pupils (typically +4 months overall). In contrast, summer schools without a clear academic component are not usually associated with learning gains, though they may have other benefits.

Behind the average

Closing the disadvantage gap

There is some evidence that pupils from disadvantaged backgrounds can benefit from summer schools, where activities are focused on well-resourced, small group or one to one academic approaches. There is less evidence for the effectiveness of light-touch interventions such as summer book sharing programmes.

Studies indicate that attendance and drop out are key challenges for voluntary, out-of-term time provision, particularly for disadvantaged pupils. To overcome these issues, schools should aim to identify any potential barriers (such as food or transport costs, clashes with planned holidays or religious events) early on, such as through communication with parents/carers to improve engagement. When targeting summer schools at pupils from disadvantaged backgrounds, approaches should seek to minimise any risk of stigmatisation.

Including additional non-academic activities such as sports, arts or cultural enrichment are valuable in their own right and can offer opportunities to pupils from low-income households who may not otherwise be able to afford them. A mix of activities may also help to promote engagement and take up of summer schools.

How could you implement in your setting?

Summer schools impact academic outcomes through providing additional time over the summer that leads to additional learning. This additional learning time may also be targeted to pupils that have struggled in particular areas of the curriculum. Schools implementing the approach should therefore consider:

Summer schools may also include other enrichment and engagement activities such as arts and sports activities or educational visits. These can be an important component for maintaining engagement in an academically targeted summer school or an important activity in and of themselves, if the summer school has broader goals.

Summer schools are typically delivered over two or three weeks. Some studies have examined longer summer school programmes of up to six weeks, though these are unusual and some have found particular issues with maintaining attendance. Schools may choose to provide programmes immediately after the end of the summer term, during the summer break, or immediately prior to the start of the new school year.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School’s Guide to Implementation](#).

What does it cost

The average cost of summer schools is moderate. The cost to schools is largely based on staff salary, facilities, resources and activity costs, the majority of which are recurring costs with some variation related to the size, duration and staffing levels of summer schools.

Summer schools will require a large amount of staff time, compared with other approaches. Summer schools may be delivered by a mix of teachers, teaching assistants, pastoral and support staff, external providers (such as literacy charities or sports groups) or volunteers.

Alongside time and cost, school leaders should consider how to maximise the academic component of summer school provision, ensuring it is well-resourced, appropriately staffed and closely targeted at pupil learning needs. School leaders should avoid approaches that increase teacher workload without securing pupil learning gains and should consider whether approaches detract from teachers' capacity to plan high-quality teaching and learning in the next academic year.

How secure is the evidence?

The security of the evidence around summer schools is rated as low. 69 studies were identified that meet the inclusion criteria for the Toolkit.

Overall, the topic lost an additional padlock because a large percentage of the studies are not randomised controlled trials. While other study designs still give important information about effectiveness of approaches, there is a risk that results are influenced by unknown factors that are not part of the intervention.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.

What is it?

Teaching assistants (also known as TAs, classroom support assistants or teachers' aides) are adults who support teachers in the classroom. Teaching assistants' duties can vary widely, but they are generally deployed in two ways; to support the teacher in the general classroom environment, or to provide targeted interventions, which are often delivered out-of-class. The role can also include administrative support.

Key findings

1. Teaching assistants can provide a large positive impact on learner outcomes, however, how they are deployed is key.

2. The high average impact hides a large variation between the different approaches to teaching assistant deployment. Targeted deployment, where teaching assistants are trained to deliver an intervention to small groups or individuals has a higher impact, whereas deployment of teaching assistants in everyday

classroom environments has not been shown to have a positive impact on learner outcomes.

3. Access to high quality teaching is the most important lever schools have to improve outcomes for their pupils. It is particularly important to ensure that when pupils are receiving support from a teaching assistant, this supplements teaching but does not reduce the amount of high-quality interactions they have with their classroom teacher both in and out-of-class.

4. Investing in professional development for teaching assistants to deliver structured interventions can be a cost-effective approach to improving learner outcomes due to the large difference in efficacy between different deployments of teaching assistants.

How effective is the approach?

The average impact of the deployment of teaching assistants is about an additional four months' progress over the course of a year. However, effects tend to vary widely between those studies where teaching assistants are deployed in everyday classroom environments, which typically do not show a positive benefit, and those where teaching assistants deliver targeted interventions to individual pupils or small groups, which on average show moderate positive benefits. The headline figure of four additional months' progress lies between these figures.

Research that examines the impact of teaching assistants deployed in everyday classroom environments suggests that students in a class with a teaching assistant present do not, on average, outperform those in one where only a teacher is present. This average finding covers a range of effects. In some cases, teachers and teaching assistants work together effectively, leading to increases in attainment. In other cases, pupils, particularly those who are low attaining or identified as having special educational needs, can perform worse in classes with teaching assistants.

Where overall negative effects have been recorded, it is likely that support from teaching assistants has substituted rather than supplemented teaching from teachers. In the most positive examples, it is likely that support and training will have been provided for both teachers and teaching assistants so that they understand how to work

together effectively, such as by making time for discussion before and after lessons.

Research which focuses on teaching assistants who provide one to one or small group targeted interventions shows a stronger positive benefit of between four and six additional months on average. Often interventions are based on a clearly specified approach which teaching assistants have been trained to deliver.

Teachers also report the benefits in terms of workload and reduced stress from working with teaching assistants.

In England, positive effects have been found in studies where teaching assistants deliver high-quality structured interventions which deliver short sessions, over a finite period, and link learning to classroom teaching, such as:

- Abracadabra (ABRA)
- Catch Up Literacy
- Catch Up Numeracy
- Nuffield Early Language Intervention (NELI)
- REACH
- Switch-on Reading
- Talk for Literacy

There is also evidence that working with teaching assistants can lead to improvements in pupils' attitudes, and also to positive effects in terms of teacher morale, workload and reduced stress.

Behind the average

Closing the disadvantage gap

Schools should carefully consider how teaching assistants are used to support learners from disadvantaged backgrounds. There is evidence that when a teaching assistant is used to support specific pupils routinely in the

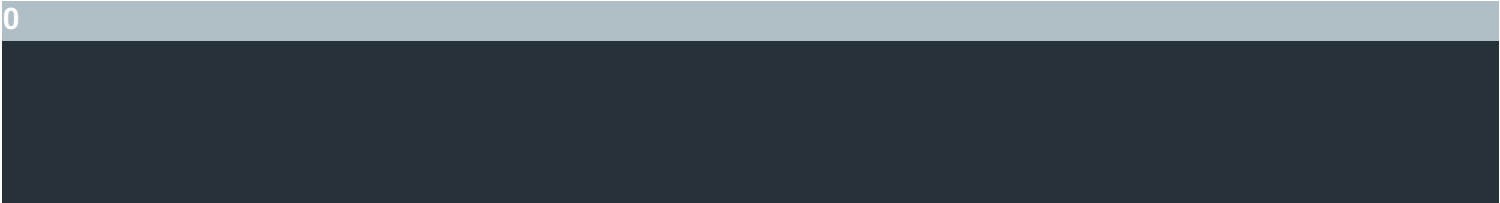
classroom, the teacher may interact less with these pupils, meaning that those who need additional teacher monitoring and support may not receive it. Therefore, additional care should be given to how teachers respond to the deployment of teaching assistants and who they are supporting, particularly for previously low attaining or disadvantaged pupils.

However, well-evidenced teaching assistant interventions can be targeted at pupils that require additional support and can help previously low attaining pupils overcome barriers to learning and ‘catch-up’ with previously higher attaining pupils.

Schools should carefully monitor teaching assistant interventions to ensure they are well-delivered, so that pupils receive the large benefits of structured interventions and not the limited impact of general deployment.

Applications and approaches

Teaching assistants in everyday classroom environments



Targeted deployment Teaching assistants delivering targeted interventions



How could you implement in your setting?

Teaching assistant interventions have an impact through providing additional support for pupils that is targeted to their needs. In order to have the desired effect schools might consider:

- Careful assessment of pupils’ needs so that teaching assistants support is well targeted
- Training teaching assistants so that interactions are high quality – for example, using well-evidenced targeted programmes
- Ensuring that any interventions are well-linked to classroom content and do not reduce high quality interactions with teachers

High quality communication between teaching assistants and classroom teachers is likely to support good implementation of teaching assistant interventions. Studies were not included where the TA was assigned as SEND support for an individual pupil as part of an SEND plan.

Teaching assistant interventions are typically delivered over half-term or term-long periods when adopting a targeted intervention or approach, or across the entire academic year when used as more general deployment.

When introducing new approaches, schools should consider implementation. For more information see [Putting Evidence to Work – A School’s Guide to Implementation](#).

What does it cost

Overall, the median costs of implementing teaching assistant interventions are estimated as moderate. The costs associated with the effective deployment of teaching assistants arise from staff salary expenses, the majority of which are recurring costs.

Whilst the median cost estimate for the effective deployment of teaching assistant interventions is moderate, differences in training and resource costs through specific teaching assistant taught programmes or interventions means that costs can range from very low to high. Evidence suggests that the more effective approaches are structured and involve high-quality support and training, so it is important that teaching assistants have professional development in the pedagogy and content of the specific intervention they are expected to use.

These cost estimates assume that schools are already paying for teacher time to work with and support teaching assistants, and the facilities and materials required to implement a teaching assistant intervention. These are all pre-requisite costs of using teaching assistant interventions, without which the cost is likely to be higher.

How secure is the evidence?

The security of the evidence around teaching assistant interventions is rated as moderate. 65 studies were identified.

Overall, the topic lost an additional padlock because a large percentage of the studies were not independently

evaluated. Evaluations conducted by organisations connected with the approach – for example, commercial providers, typically have larger impacts, which may influence the overall impact.

What is it?

Within-class grouping (also known as within-class attainment grouping) means organising pupils within their usual class for specific activities or topics, such as literacy or mathematics. Pupils with similar levels of current attainment are grouped together, for example, on specific tables, but all pupils are taught by their usual teacher and support staff, and they usually all follow the same curriculum but at different levels of difficulty.

The aim of this type of grouping is to match tasks, activities and support to pupils' current capabilities, so that all pupils have an appropriate level of challenge.

Within-class grouping can involve the use of other approaches such as [collaborative learning](#) or targeted strategies (see [reading comprehension strategies](#)).

Although within-class grouping is sometimes described as 'ability grouping', we refer here to 'attainment' rather than 'ability', as schools generally use measures of current performance, rather than measures of ability, to group pupils.

In the UK, within-class attainment grouping tends to be more common in primary schools than in secondary schools. In primary schools this may involve pupils grouped together on specific tables, but this is typically not the case in secondary schools. The research evidence does not examine the effect of pupils working in groups, but rather the effect of pupils being grouped by attainment which may, or may not, involve them working in groups.

Key findings

1. Within class attainment grouping has a positive impact, on average, of 2 months additional progress. The [evidence strength](#), however, is very limited and there is variation behind this average.

2. It is important to carefully consider which content is appropriate for within class attainment grouping. While the

impact in maths was positive, studies that measured literacy outcomes found no difference, on average.

3. Consider the impact of within class attainment grouping on pupils with low prior attainment and carefully monitor engagement and attitudes to learning.

4. One advantage of within class grouping might be flexibility in grouping arrangements. Pupils progress at different rates so regular monitoring and assessment is important to minimise misallocation and ensure challenge for all pupils.

How effective is the approach?

The average impact of the within-class grouping is about an additional two months progress over the course of a year.

Within-class attainment grouping may also have an impact on wider outcomes such as confidence. Some studies from the broader evidence base conclude that grouping pupils on the basis of attainment may have longer term negative effects on the attitudes and engagement of low attaining pupils, for example, by discouraging the belief that their attainment can be improved through effort.

Behind the average

Closing the disadvantage gap

As with setting and streaming, evidence suggests that disadvantaged pupils can suffer from lower teacher expectations which increases their chances of being grouped with lower prior attainers. If groups are not arranged flexibly, and the impact of grouping on pupil engagement and motivation not monitored, within-class grouping could have negative effects for disadvantaged pupils.

One recent study commissioned by the EEF found that schools with higher proportions of disadvantaged pupils seem to be more likely to use whole-class teaching with KS2 pupils, as opposed to within-class grouping. This finding was not consistent across Key Stages however.

How could you implement in your setting?

By adapting teaching to pupils' needs and prior knowledge, teachers may be able to support, stretch, and challenge pupils' learning more effectively.

Effective implementation of within class grouping approaches might include:

The evidence indicates higher impacts, on average, for mathematics. It is particularly important that teachers consider appropriate content for within class attainment grouping.

Within-class grouping interventions may be used as often as teachers require in their daily practice. Within-class groupings might be used as a temporary process for specific tasks, or a more regular routine in which seating plans are orientated around prior attainment outcomes. This latter approach should be implemented with caution, as without flexibility to move between groups some pupils may suffer from a lack of confidence leading to lower engagement, and subsequently attainment.

What does it cost

Overall, the median costs of implementing within-class grouping are estimated as very low. The costs associated with within-class grouping arise from the production of any additional resources (e.g., scaffolds or prompt sheets) that are provided to groups with differing levels of prior attainment.

This cost estimates assume that schools are already paying for teacher time, assessment methods, and potentially IT software to monitor pupil needs and attainment. These are all pre-requisite costs of implementing within-class grouping, without which the cost is likely to be higher.

How secure is the evidence?

The security of the evidence around within class attainment grouping is rated as very low. Only 22 studies were identified that meet the inclusion criteria of the Toolkit. The topic lost an additional padlock because a small percentage of studies have taken place recently. This might mean that the research is not representative of current practice.

As with any evidence review, the Toolkit summarises the average impact of approaches when researched in academic studies. It is important to consider your context and apply your professional judgement when implementing an approach in your setting.